The Macroeconomic Effects of Global Supply Chain Shocks

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Abstract

This paper provides evidence that global supply chain shocks are key drivers of business cycle fluctuations, introducing a novel identification strategy based on a narrative analysis of price surcharges from the three largest container shipping companies. Negative shocks cause a persistent rise in consumer prices and a prolonged decline in economic activity. Sectoral impacts vary with exposure to global supply chains, proxied by the import share of inputs. Spillovers extend to non-tradable sectors. These shocks accounted for up to 45% of the post-pandemic inflation. In a counterfactual with only global supply chain shocks, recovery would have taken 18 months longer.

Keywords: Business Cycles, Global Supply Chain Shocks, Supply Chain Disruptions, Narrative Identification, Post-Pandemic Inflation

JEL Codes: E31, E32, F14, R41

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1 Introduction

The surge in post-pandemic inflation in the United States has sparked an intense debate over its primary causes. Several studies attribute the majority of the price surge to demand-side forces, such as pent-up consumption and expansionary fiscal policy, while others point to supply-side factors, including energy price shocks and production bottlenecks. Despite disagreeing on the relative importance of demand versus supply forces, both academics and policymakers uniformly acknowledge global supply chain disruptions as key contributors to the post-pandemic inflation.

Yet isolating an exogenous measure of global supply chain shocks, and with it a clean estimate of their macroeconomic consequences, remains an open challenge. The difficulty is fundamentally one of identification, arising from the inherently global and heterogeneous nature of these shocks. Global supply chain disruptions can be caused by port equipment failures, labor strikes, extreme weather events, and piracy, which also makes them difficult to measure consistently. Lacking a direct measure, researchers rely on indicators such as delivery times, port congestion, and shipping costs, but these are themselves endogenous to fluctuations in global demand and production activity. Consequently, the extent to which shocks to the global supply chain drive business cycle fluctuations remains an open question.

Macroeconomic theory provides clear reasons why shocks to the global supply chain might have large aggregate effects. Acemoglu et al. (2012) show that when a central supplier provides inputs to a disproportionately large number of downstream industries, idiosyncratic shocks to that sector can generate significant aggregate fluctuations. The global supply chain plays precisely this role: between 80% and 90% of internationally traded goods travel by sea, and roughly two-thirds—worth over $8 trillion—move in containers (Notteboom, Pallis and Rodrigue, 2022). Moreover, a handful of global carriers dominate the market and have fused the world's major trade lanes—East–West, Trans-Pacific, and Trans-Atlantic—into a single, tightly coupled global supply chain by linking vessels through key global ports. In line with the granularity argument of Gabaix (2011), idiosyncratic shocks to these dominant players or key ports can generate aggregate fluctuations. Recent research from Barrot and Sauvagnat (2016), Acemoglu and Tahbaz-Salehi (2020, 2025), Elliott, Golub and Leduc (2022) and Elliott and Golub (2022) shows that complex production networks in equilibrium are inherently fragile to small shocks that simultaneously disrupt multiple supply relations. Because the global supply chain physically intermediates all the other supply chains, localized shocks can cascade through it, disrupting supply relations across the entire global economy.

This paper makes five contributions. (i) I propose a new strategy to identify exogenous shocks to the global supply chain. (ii) I show that small shocks to the global supply

chain generate significant aggregate fluctuations in the US economy. (iii) I find that these shocks produce stagflationary effects across all tradable sectors and spill over to non-tradable services. (iv) I contribute to the debate about the causes of the post-pandemic inflation, concluding that these shocks were key contributors to inflation in 2021 and exerted substantial downward pressure on the economic recovery. (v) Lastly, I show that shocks to the global supply chain account for a significant share of the variability in macroeconomic activity over the business cycle.

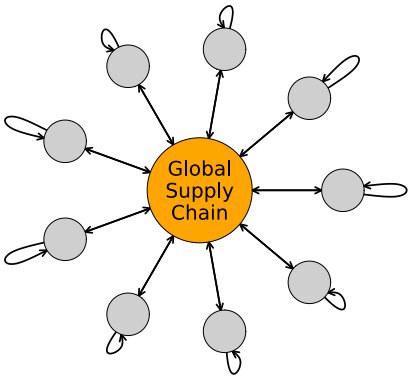


Figure 1: The global supply chain as the central hub of the global economy

Notes: The figure provides a stylized representation of the global economy as a star network. Each peripheral node represents an economy that maintains both domestic (self-loops) and international supply relations. All international supply chains are mediated by the global supply chain, which functions as the central hub of the network.

Two lines of empirical work precede this paper. The first builds aggregate measures of supply chain pressure and estimates their effects directly. Benigno et al. (2022) construct the Global Supply Chain Pressure Index and trace its inflation pass-through with local projections; Bai et al. (Forthcoming) build an Average Congestion Rate index from container-ship data and identify disruptions through sign restrictions from a structural model of goods-market congestion. Related work embeds such measures in SVARs identified through recursive, sign, or narrative-sign restrictions (Finck and Tillmann, 2022; Carrière-Swallow et al., 2023; Gordon and Clark, 2023; De Santis, 2024; Ascari, Bonam and Smadu, 2024). Yet key variables such as commodity prices, delivery times, shipping costs, and port congestion are endogenously influenced by global economic activity (Baumeister and Hamilton, 2019; Alquist, Bhattarai and Coibion, 2020; Delle Chiaie, Ferrara and Giannone, 2022; Baumeister, Korobilis and Lee, 2022). This complicates causal inference, as these indicators conflate the effects of supply chain disruptions with those of other macroeconomic shocks. The second exploits cross-sectional variation around exogenous events such as the Tōhoku earthquake

(Boehm, Flaaen and Pandalai-Nayar, 2019; Carvalho et al., 2021), achieving credible identification but tying its conclusions to particular episodes.

I complement both strands of the literature by developing a new strategy to identify global supply chain shocks based on narrative analysis of price surcharge announcements from the three largest containerized shipping companies, which collectively hold nearly 50% of the market share and up to 70% through strategic alliances. I collected 7,872 price and surcharge updates published on their websites from 2014 to 2024, focusing exclusively on five key regions: Europe, the Middle East and Indian Subcontinent, East Asia, Oceania, and North America. Most of these updates are routine adjustments—general rate increases, peak-season surcharges, and port additionals—through which these oligopolistic carriers continuously reprice in response to shifting demand and capacity conditions. I isolate the price surcharges that carriers tie to unpredictable, supply-side disruptions using a keyword-based algorithm adapted from Baker, Bloom and Davis (2016), which flags 369 announcements that contain terms objectively associated with exogenous supply-side events, such as strikes, natural disasters, armed conflicts, piracy, and operational disruptions. I then conduct a manual, event-by-event narrative analysis of each of these 369 candidates, cross-validating the companies' stated rationale against independent external sources such as local and international newspapers, port authority reports, and contemporaneous news coverage. Events are retained only when this narrative analysis confirms they are exogenous, unpredictable, and unrelated to other shocks that could independently affect the variables of interest. This procedure is an adaptation of the narrative identification strategy widely employed in the study of oil shocks (Hamilton, 1985, 2003; Caldara, Cavallo and Iacoviello, 2019; Qureshi and Ahmad, 2025), monetary policy shocks (Romer and Romer, 2004; Aruoba and Drechsel, Forthcoming), fiscal shocks (Ramey and Shapiro, 1998; Romer and Romer, 2010; Ramey, 2011), and tariff shocks (den Besten et al., 2026). This process allows me to identify 54 exogenous supply chain disruptions, measured as price surcharges in US dollars per Forty-foot Equivalent Unit (FEU) dry container.

One central concern is that, while the timing of each disruption is exogenous, the size of a surcharge could in principle reflect prevailing demand conditions rather than the disruption itself. Two findings speak against this. An indicator instrument that retains only the sign and timing of each event, discarding its magnitude, delivers essentially the same baseline results. Demand conditions along the supply chain also do not Granger-cause the instrument, as they would if surcharge magnitudes encoded demand. For China, the Euro Area, Japan, S. Korea, Taiwan, the United Kingdom, and the United States, I study the predictive power of the two demand subcomponents of the manufacturing PMI, new orders and quantities purchased, capturing customer demand for firms' products and firms' demand for intermediate

inputs, together with demand-influenced conditions such as delivery times, purchased stocks, input prices, and backlogs. Both results are corroborated by carriers repricing to demand continuously. General rate increases and peak-season surcharges arrive at roughly one per business day (2,794 over 2014–2024, 35.5% of all announcements), against the 54 disruption events I retain. Because demand is repriced in real time through these dedicated channels, it is not withheld and released under cover of a supply-side disruption. The surcharges I isolate sit separately, on top of this routine repricing.

One might worry that some events reach the economy through channels other than the global supply chain. Those at the Strait of Hormuz, for instance, could reflect oil shocks. I find, however, that the instrument is uncorrelated with available oil-shock measures, that controlling for an identified oil shock leaves the estimated responses unchanged, and that the price of oil never responds significantly to my instrument in any specification. Events at US ports, in turn, could affect the US economy through a direct domestic channel, threatening the exclusion restriction; excluding the five such events leaves the baseline results identical. I address these concerns, together with additional robustness checks, in the sensitivity analysis of Section 6.

Using the FEU price surcharges as an external instrument within a Proxy-SVAR framework, I estimate the dynamic causal effects of shocks to the global supply chain on the US economy. I find that negative shocks produce aggregate stagflationary effects, with a persistent rise in consumer prices and a prolonged economic slowdown marked by lower industrial production and higher unemployment. At the sectoral level, I find declines across all production sectors, with sustained increases in all consumer price categories. Sectoral output dynamics mirror aggregate responses: an initial decline, peaking within the first few quarters, and a gradual return to steady state over three years. However, the magnitude of these effects varies across sectors. A sector's exposure to the global supply chain, proxied by the share of its manufactured inputs sourced from abroad, directly or indirectly through its suppliers' own suppliers (Baldwin, Freeman and Theodorakopoulos, 2023), explains this heterogeneity. I find that greater exposure is associated with more severe production declines. Moreover, sectors that experience the largest output reductions also exhibit the most significant and persistent pass-through to consumer prices. I also document significant spillover effects into non-tradable service sectors. Although services are not directly exposed to the global supply chain, service activity and employment decline with a lag.

I also contribute to the debate about the main causes of the post-pandemic inflation. My estimates indicate that shocks to the global supply chain were key drivers of inflation in 2021, accounting for an average of 45% of the price increase. While my analysis does

not rule out the significant role of demand,[1] it shows that these shocks were the primary contributors in the early stage of the price surge in 2021. While from 2022 onward other factors, such as demand forces, fiscal stimulus, and other supply shocks,[2] increasingly shaped inflation dynamics, global supply chain shocks remained a significant force, contributing, according to my estimates, 29%, 28%, and 21% to consumer price increases in 2022, 2023, and 2024, respectively. Regarding economic recovery, these shocks exerted sustained downward pressure. In a counterfactual scenario with only global supply chain shocks, industrial production would have required an additional 18 months to return to pre-pandemic levels, from November 2021 to May 2023.

I show more broadly that shocks to the global supply chain are key drivers of business cycle fluctuations. They explain 30% of consumer price variability, roughly 25% of industrial production variability, and about 10% of unemployment variability over the business cycle.

The closest contribution to this study is Känzig and Raghavan (2026), developed independently and circulated after the first version of this paper. They construct a measure of supply chain disruptions from daily changes in the Baltic Exchange Dry Index around exogenous events at the Panama and Suez Canals. The two approaches are complementary and capture different segments of the supply chain: their instrument reflects shocks to dry-bulk shipping identified at these chokepoints, whereas mine captures disruptions to high-value containerized intermediate and final goods across all major trade corridors, including the Trans-Pacific, Trans-Atlantic, and key Asian hubs. Our results agree on the aggregate effects, but my broader coverage shows that these shocks account for a substantial share of business-cycle fluctuations. The same distinction shapes the sectoral results: a dry-bulk measure concentrates pass-through in commodity-intensive sectors, whereas the containerized coverage traces broad-based pass-through across intermediate and final goods.

The paper is organized as follows: Section 2 discusses the identification design and the construction of the instrument. Section 3 outlines the econometric approach and examines the propagation of global supply chain shocks at aggregate and sectoral levels. Section 4 evaluates their cumulative impact on pandemic-era inflation and economic recovery. Section 5 assesses the quantitative role of global supply chain shocks as drivers of business cycle fluctuations. Section 6 performs some robustness checks, and Section 7 concludes.

[1] Di Giovanni et al. (2022), Giannone and Primiceri (2024), Garcia Revelo, Levieuge and Sahuc (Forthcoming), and Bergholt et al. (Forthcoming) attribute the majority of the price surge to strong pent-up demand, while Cochrane (2022), Di Giovanni et al. (2023*b*,*a*), Jordà and Nechio (2023), Faria-e Castro (2024), and Hazell and Hobler (2024) emphasize the inflationary impact of fiscal stimulus.

[2] On the supply side, Bernanke and Blanchard (2023, 2024) identify shocks to energy and food prices, alongside widespread supply disruptions in key sectors, as key inflationary drivers. Caldara, Iacoviello and Yu (2025) argue that supply forces primarily drove inflation, Comin, Johnson and Jones (2023) find that potentially-binding capacity constraints explain half of the rise in inflation, while Goda and Soltas (2022), Hobijn and Sahin (2022), Lee, Park and Shin (2023) discuss the role of potential shocks to productive capacity.

2 Narrative Identification

The identification strategy builds on the narrative analysis of surcharge announcements from the world's three largest shipping companies. When confronted with exogenous disruptions that increase operational costs, shipping companies respond by imposing price surcharges on shippers. The idea is to construct a series of price surcharges correlated exclusively with exogenous supply-side disruptions that will serve as an external instrument in a SVAR framework to structurally identify the unobserved global supply chain shocks. Before detailing the procedure for identifying such disruptions, I first provide evidence on the containerized shipping market's oligopolistic structure and the role played by price surcharges in the measurement and transmission of shocks across the supply chain network.

2.1 The Containerized Shipping Industry

From 2014 to 2024, the containerized shipping industry was dominated by three major alliances: the 2M Alliance, formed by Mediterranean Shipping Company (MSC) and A.P. Møller–Mærsk A/S (Maersk); the Ocean Alliance, comprising Compagnie Maritime d'Affrètement – Compagnie Générale Maritime (CMA CGM), COSCO, and Evergreen; and THE Alliance, including Hapag-Lloyd, Ocean Network Express (ONE), Yang Ming Marine Transport Corporation, and Hyundai Merchant Marine (HMM). These alliances collectively controlled approximately 80-90% of global container shipping capacity by the early 2020s, reflecting the concentrated structure of the industry and enabling carriers to coordinate vessel deployment, share terminal infrastructure, and reduce route duplication, thereby achieving significant economies of scale. High fixed costs, substantial barriers to entry, and the need for globally integrated service networks reinforced market power among a small group of dominant carriers. The three largest shipping companies, MSC, Maersk, and CMA CGM, together account for nearly 50% of the global container shipping market, with operations that span all major trade corridors, from Asia to Europe, North America, and beyond.

Crucially, the interconnectedness of major trade lanes, including the East-West, Trans-Pacific, and Trans-Atlantic corridors, effectively creates a de facto single global shipping route. Vessels routinely interconnect at major hubs such as Singapore, Shanghai, Rotterdam, and Los Angeles–Long Beach, transforming the global network into an integrated and interdependent system (Fan, Wilson and Dahl, 2012). As a result, disruptions in one region, such as the 2021 Suez Canal blockage or the 2022 Shanghai port backlog, quickly reverberated across the entire network, generating ripple effects on capacity, schedules, and freight pricing worldwide.

Within this tightly interconnected system, shipping companies respond to demand or

supply shocks by imposing price surcharges on shippers. These surcharges serve as a key mechanism to recover rising operational costs and, at the same time, to rebalance container flows by disincentivizing shipments into already congested or container-saturated regions. However, because surcharges are typically applied across entire trade lanes, a localized disruption, originating in, say, a single port or country, can lead to system-wide price increases, affecting routes far removed from the original shock. In this way, price surcharges function as transmission vehicles: they not only quantify the magnitude of the disruptions but also propagate the increasing operational costs across multiple trade lanes through the combined forces of oligopolistic power and network interdependence. MSC, Maersk, and CMA CGM published more than 7,000 price and surcharge updates on their websites between January 2014 and December 2024, reflecting changes in container shipping costs across a wide range of global routes and ports. I will now describe the methodology used to isolate exogenous supply-side disruptions to the global supply chain from the narrative record of price surcharges.

2.2 Identifying Exogenous Disruptions

Methodology.— To construct a narrative-based measure of exogenous global supply chain disruptions, I begin by systematically collecting price surcharge announcements from the websites of the three largest container shipping companies. Specifically, I compile 1,217 surcharges from MSC (January 2014–December 2024), 2,713 from Maersk (March 2019–December 2024), and 3,942 from CMA CGM (January 2014–December 2024). These announcements include various surcharge types, often accompanied by textual justifications provided by the carriers themselves. Most are routine commercial adjustments through which these oligopolistic carriers continuously reprice in response to shifting market conditions. To quantify this, I classify each announcement by its stated surcharge type, usually named in the title; when the title carries only the trade lane, I read the opening of the text instead. Peak-season surcharges, unambiguously a response to demand, account for 996 announcements (12.7%); general rate changes, which respond jointly to demand and supply conditions, account for another 1,798 (22.8%). Routine repricing therefore amounts to 2,794 of the 7,872 announcements (35.5%). The count is a lower bound, since announcements go unclassified when neither the title nor the opening names a charge. For CMA CGM, which names the type almost without exception, the share rises to 47.2% (see Online Appendix Section E.1 for the complete decomposition). Only a minority are extraordinary surcharges triggered by exogenous, supply-side disruptions. Isolating this smaller set is the goal of the identification procedure I describe next.

I focus exclusively on surcharges related to the three major trade routes, Trans-Pacific, East-West and Trans-Atlantic corridors. I exclude routes serving South America and West Africa, as well as surcharges involving ports with annual throughput below 250,000 FEUs, which are less relevant for capturing globally systemic disruptions.

To isolate surcharges only related to exogenous supply-side disruptions, I proceed in two steps. First, I adopt a keyword-based algorithm adapted from Baker, Bloom and Davis (2016). Using a pre-defined list of terms associated with operational failures, labor strikes, natural disasters, geopolitical conflicts, and partial or full blockages of maritime chokepoints with consequent re-routing (see Online Appendix Section E.2), I isolate a subset of 369 surcharge announcements that contain one or more terms plausibly linked to unpredictable supply-side disruptions.

Second, I conduct a manual, event-by-event narrative analysis of each of the 369 candidate surcharge announcements. For every event, I cross-validate the carrier's stated rationale against independent external sources, local and international newspapers, contemporaneous news coverage, port authority reports, and industry publications, to confirm that the surcharge reflects an exogenous, supply-side disruption unrelated to broader macroeconomic conditions, demand forces, or other structural shocks. Any announcement for which this confirmation cannot be established is excluded from the proxy. Of the initial 369 candidates, only 54 survive this screening. The full narrative analysis is detailed in the Online Appendix, where each retained event is documented following a common structure: the technical details of the surcharge, the background conditions at the affected ports, and the independent evidence identifying the cause of the surcharge.

Applying the criteria.— For illustration, I include below a price surcharge announcement published by Maersk on January 8, 2024, which suspended transit through the Red Sea due to multiple Houthi rebel attacks on commercial vessels, rerouted vessels via the Cape of Good Hope, and introduced a Transit Disruption Surcharge (TDS):

> ***08 January 2024 – Transit Disruption Surcharge***
>
> *As we have been communicating since mid-December/23, Maersk is continuing monitoring developments around the Red Sea / Gulf of Aden and making carefully considered changes to services to ensure the safety of our seafarers, vessels and customers' cargo.*
>
> *The situation is constantly evolving and remains highly volatile, and all available intelligence at hand confirms that the security risk continues to be at a significantly elevated level. We have therefore decided that all Maersk vessels due to*

transit the Red Sea / Gulf of Aden will be diverted south around the Cape of Good Hope for the foreseeable future.

By suspending voyages through the Red Sea / Gulf of Aden, we hope to bring our customers more consistency and predictability despite the associated delays that come with the re-routing.

The previously announced surcharges for all cargo on vessels affected by the disruptions around the Red Sea / Gulf of Aden remain in effect:

Transit Disruption Surcharge effective as from actual sailing date 10/01/2024 until further notice.

To give you as much clarity and predictability as possible, TDS will apply only to bookings confirmed as from 21/12/2023 (global announcement date) and departure date as from 10/01/2024 onwards.

TDS - USD per container	**20DRY**	**40DRY**	**40HREF**
All shipments diverted from Suez	200	400	450

The surcharge announcement itself makes the exogenous supply-side cause transparent, but even in such cases, I cross-validate the carrier's rationale against independent sources before retaining the event. In other cases, the carrier provides no explicit justification, and the external sources become the primary basis for classification. For example, CMA CGM introduced a Port Congestion Surcharge (PCS) on March 27, 2023, for shipments departing from Mersin, Turkey:

27 March 2023 – PCS in Mersin, Turkey

We inform our customers that we are experiencing important delays in Mersin, Turkey between congestion/waiting time plus berth time on the current vessels berthing sequences with following impacts: (i) laden cargo lead-time is being increased; (ii) empty equipment re-loading / repositioning frequency is affected; (iii) operating costs are increased.

In this respect please be informed that CMA CGM will be implementing a Port Congestion Surcharge in Mersin as follows:

- ***From:*** *Mersin, Turkey*
- ***To:*** *Mediterranean & Adriatic, Black Sea, North Africa & Portugal, South America, Central America & Caribbean, Mexico West Coast, Windward,*

Guyana North Brazil, Middle East Gulf, Red Sea, North & South East Asia, China, Hong Kong SAR

- ***Cargo:*** *Dry*
- ***Applicable from:*** *March 27th, 2023, loading date (April 25th, 2023 for Puerto Rico)*
- ***Amount:*** *USD 100 per TEU*

Although CMA CGM did not specify the reason for the PCS, I classify this surcharge as a response to an exogenous supply-side disruption. Several sources confirmed that the cause of the surcharge was a devastating 7.8-magnitude earthquake that struck southern Turkey in February 2023, causing extensive damage to port infrastructure and regional road networks.[3] Trucking activity around Mersin was largely disrupted, and the major nearby port of Iskenderun ceased operations due to fire damage caused by the earthquake. As a result, vessels were rerouted to Mersin, leading to increased delays and congestion.

Morphology of exogenous disruptions.— Overall, the narrative analysis leads to the identification of 54 exogenous disruptions. Among these, 10 events involve major maritime chokepoints, where disruptions affect all trade routes transiting through them. These include three weather-related disruptions at the Panama Canal, driven by prolonged drought and reduced water levels; four operational disruptions at the Suez Canal, caused by the 2021 grounding of the Ever Given and other blockage-related incidents; one conflict-related disruption in the Strait of Hormuz, linked to an Iranian attack on two oil tankers; and two war-related events at the Bab el-Mandeb Strait, both stemming from Houthi rebel attacks and the broader civil conflict in Yemen. Beyond chokepoints, 7 additional events involve all ports along major trade corridors. These include disruptions on routes connecting Far East Asia (China, Taiwan, Hong Kong) to the United States and Europe, as well as six events along the Trans-Atlantic corridor, primarily caused by low water levels in the St. Laurent Seaway. An additional 8 disruptions affect broader macro-regions, involving multiple ports or inland transport systems within the same country or trade area. These include, among others, widespread disruptions across dry ports in the United States; strikes in southern France and nationwide in Spain; operational challenges across all UK ports; and tropical storm and monsoon flooding disrupting all Myanmar ports. Finally, 29 events are localized at individual ports, including high-traffic terminals such as Shanghai, Ningbo, Los Angeles–Long

[3]See: Portcast (2023): *Impact of Türkiye–Syria Earthquake on the Shipping Industry*, Fox Business (2023): *Earthquake Sparks Massive Shipping Container Fire*, and Reuters (2023): *Iskenderun Port Operations Halted After Earthquake-Induced Fire*.

Beach, Felixstowe, Nhava Sheva, and Colombo, affected by either operational failures or strikes. Notably, only five of the 54 disruptions occur at US ports.

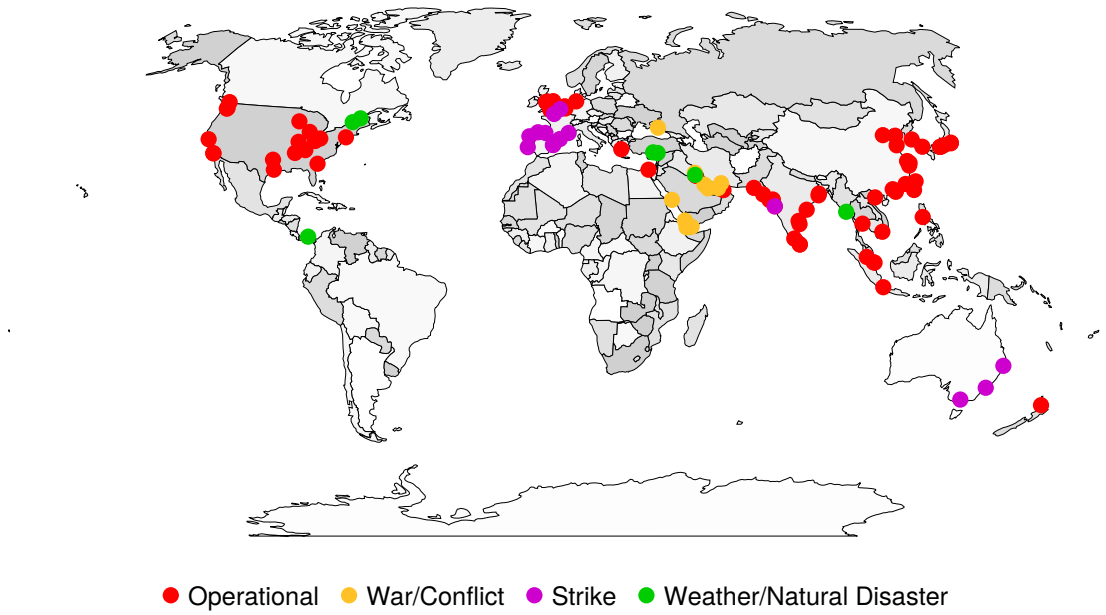


Figure 2: Locations of seaports and dry ports affected by exogenous disruptions.

Notes: The figure shows the geographic location of the identified exogenous supply chain disruptions, color-coded by underlying cause: operational, war/conflict, strike, and weather/natural disaster. For events affecting an entire trade route, only the largest ports in terms of annual dry FEU container throughput are shown to avoid overcrowding the visual.

These disruptions are classified into four categories based on their underlying cause: operational, war and conflict, strikes, and extreme weather or natural disasters. Figure 2 illustrates the global distribution of identified supply chain disruptions, with each event categorized by its underlying cause. For disruptions affecting entire trade routes, only the busiest ports, measured by annual container throughput in FEUs, are shown to preserve visual clarity.

Operational disruptions, comprising just above 40% of the total (22 events), include equipment and automation failures, such as the port-crane breakdowns at Shuwaikh, the failure of the automated straddle carriers at Auckland, the work refusal triggered by the Pier 400 automation rollout at Los Angeles-Long Beach, road and bridge infrastructure breakdowns, and full or partial blockages of strategic chokepoints, in particular the Suez Canal Ever Given grounding among others. War and conflict-related disruptions account for 19% of events (10 events) and involve port closures due to hostilities, particularly in the Middle East, and suspensions of trade routes in war zones, such as the Black Sea. Strikes represent 15% (8 events), comprising labor actions that affected port operations either locally

(Nhava Sheva, Lisbon, Bilbao, Le Havre) or at the country/regional level (Spain, southern France, Australia, northern Europe). Finally, natural disasters and extreme weather events constitute the remaining 26% (14 events), including the prolonged drought and El Niño-induced low water levels at the Panama Canal, the low water levels in the St. Laurent Seaway, the Turkish earthquake of February 2023 affecting Mersin and Iskenderun, and the Asian monsoon flooding in Myanmar. An overview of all surcharge dates and data sources can be found in Online Appendix Section E.2.

2.3 Construction of the Instrument

Size and timing.— Building on the previously identified exogenous disruptions and their associated price surcharges, I now proceed with the construction of the narrative-based time series. For each disruption event, I first compute the average surcharge across affected trade lanes and shipping companies. These event-level averages are then aggregated within the month in which the surcharges were implemented, producing a cumulative monthly surcharge series measured in US dollars per FEU container.

In most cases, surcharges are applied either immediately or within the same month as the corresponding announcement. When implementation occurs in the following month, I assign the surcharge to the month of the implementation. This choice is motivated by several factors. Unlike central banks or institutions such as OPEC, shipping companies communicate pricing decisions less clearly and with limited coordination. Moreover, there is no evidence that market participants actively monitor or react to these announcements in real time. Even if such monitoring were to occur, substituting containerized ocean freight with alternative modes of transport, such as air freight, is both costly and logistically challenging. In addition, the contractual terms of container shipping are typically fixed well in advance, often for periods exceeding one year, further limiting the scope for anticipatory adjustments. These features imply that the demand for containerized shipping is relatively inelastic in the short run and news effects are minimal. In the few cases where different carriers issue surcharges related to a single event across two consecutive months, I assign the event to the later month if most implementations occur after the midpoint of the earlier month.

The price surcharge series.— Figure 3 displays the resulting time series of price surcharges, expressed in US dollars per FEU dry container. This series represents a partial but timely measure of cumulative, exogenous supply-side disruptions in the global shipping network. Constructed at a monthly frequency over eleven years, it contains 43 non-zero observations corresponding to 54 individually identified disruption events. Of these non-zero

observations, 17 occurred before January 2020. It is worth noting that price surcharge data from Maersk are only available from March 2019 onward.

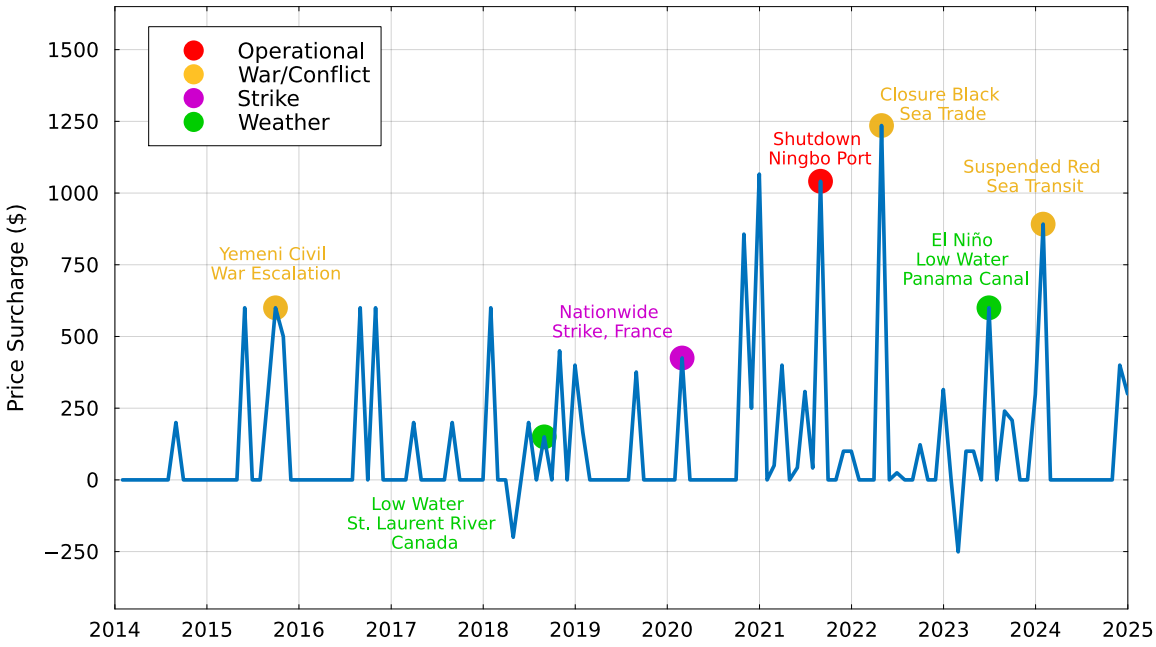


Figure 3: The Dry FEU Price Surcharge Series.

Notes: The figure shows the monthly dry FEU surcharge series, expressed in USD per container, constructed from the 54 identified exogenous disruptions. A selected subset of high-profile disruption events is also annotated and color-coded by underlying cause.

Among the identified exogenous price surcharges, only two negative values appear, each reflecting an unexpected and exogenous improvement in shipping conditions. The first is the partial removal of the Aden congestion surcharge in April 2018, triggered by a UN-led conference in Geneva that suddenly secured $2 billion in humanitarian aid for Yemen and introduced a coordinated plan to restore port logistics. The second is the reduction in inland transportation costs across India's dry port network in February 2023, caused by the surprise early commissioning of a new section of the high-capacity Eastern Dedicated Freight Corridor.

Overall, the FEU price surcharge series closely aligns with major narrative episodes of disruption. Figure 3 highlights several major events in the container shipping network. These annotated episodes span each of the four disruption categories defined earlier: geopolitical conflict (e.g., the Yemeni civil war and Black Sea trade restrictions), operational failures (the August 2021 sudden shutdown of Ningbo's Meishan terminal due to a single dockworker testing positive for COVID), extreme weather events (e.g., El Niño-related congestion in the Panama Canal and low water levels in the St. Laurent River), and labor strikes (e.g., the nationwide strike in France affecting all forms of transport).

Diagnostics.— Following the approach outlined in Ramey (2016), I perform a series of diagnostic checks on the FEU price surcharge series, examining serial correlation, Granger causality with respect to a large set of macroeconomic variables, demand-driven shipping and supply-chain conditions, and correlation with other structural shocks. The series satisfies all the properties required of a valid external instrument: it exhibits no serial correlation, it cannot be forecast by standard macroeconomic indicators, nor by demand-driven shipping and supply-chain conditions, and it is orthogonal to other structural shocks in the literature. The full results are discussed in detail in Section 6, where I also assess the validity of the identification strategy and the robustness of the empirical results more broadly.

3 Global Supply Chain Shocks & Business Cycle

The FEU price surcharge series can be seen as a partial measure of the unobserved global supply chain shock. Provided that this narrative-based series is correlated with the global supply chain shock and uncorrelated with other structural shocks affecting the economy, I use it as an external instrument within a Vector Autoregression (VAR) framework. This approach, also known as a Proxy-SVAR or SVAR-IV, focuses exclusively on identifying the structural shock of interest and builds on the extensive literature on the identification of structural VARs using external sources of exogenous variation, as developed by Stock (2008), Mertens and Ravn (2013), Stock and Watson (2018), and Montiel Olea, Stock and Watson (2021).

3.1 Econometric Framework

Consider the stationary reduced-form VAR(p) model:

$$\boldsymbol{y}_t = \boldsymbol{\Pi}\boldsymbol{x}_{t-1} + \boldsymbol{\epsilon}_t. \tag{1}$$

The k x 1 vector $\boldsymbol{x}_{t-1} = (\mathbf{1}, \boldsymbol{y}'_{t-1}, \ldots, \boldsymbol{y}'_{t-p})'$ contains an intercept and p lags of each of the variables ($k = 1 + pn$) and $\boldsymbol{\epsilon}_t$ is an n x 1 vector of reduced form innovations such that $\boldsymbol{\epsilon}_t \sim (\mathbf{0}, \boldsymbol{\Omega})$. The n x k matrix of reduced-form coefficients $\boldsymbol{\Pi}$ is estimated by OLS. I now assume that the model is invertible, i.e. the reduced-form innovations are a linear combination of the structural shocks:

$$\boldsymbol{\epsilon}_t = \boldsymbol{A}^{-1}\boldsymbol{u}_t. \tag{2}$$

where $\boldsymbol{A^{-1}}$ is the n x n structural impact matrix and $\boldsymbol{u}_t$ is the n x 1 vector of mutually uncorrelated structural shocks, i.e. $\boldsymbol{u}_t \sim (\boldsymbol{0}, \boldsymbol{D})$, where $\boldsymbol{D} = diag(d_{11} \dots d_{nn})$. In Section 6, I formally test the invertibility assumption using the procedures outlined in Plagborg-Møller and Wolf (2022) and Hamilton (Forthcoming), and fail to reject the null hypothesis of invertibility for the baseline empirical model. The invertibility assumption implies

$$\boldsymbol{\Omega} = \boldsymbol{A^{-1}} \boldsymbol{D} (\boldsymbol{A^{-1}})'. \tag{3}$$

Since the VAR is stationary, (1) yields a VMA(∞) representation of the form

$$\boldsymbol{y}_t = \sum_{h=0}^{\infty} \boldsymbol{\Psi}_h \boldsymbol{\epsilon}_{t-h}. \tag{4}$$

where the weights $\boldsymbol{\Psi}_h$ are functions of the reduced-form coefficient matrix $\boldsymbol{\Pi}$. Given the invertibility assumption, I obtain the structural vector moving average representation

$$\boldsymbol{y}_t = \sum_{h=0}^{\infty} \boldsymbol{\Psi}_h \boldsymbol{A^{-1}} \boldsymbol{A}\, \boldsymbol{\epsilon}_{t-h} = \sum_{h=0}^{\infty} \boldsymbol{\Theta}_h \boldsymbol{u}_{t-h}. \tag{5}$$

where $\boldsymbol{\Theta}_h = \boldsymbol{\Psi}_h \boldsymbol{A^{-1}}$ is the n x n matrix of structural responses to the vector of structural shocks that happened h periods before. Without loss of generality, I order the global supply chain shock as the first element (u_{1t}) of the vector of structural shocks $\boldsymbol{u}_t$; therefore, I aim to identify the first column of $\boldsymbol{A^{-1}}$, denoted by the vector $\boldsymbol{h}_1$.

Inference via External Instrument.— The identification strategy consists of using the stationary series of FEU price surcharges (Z_t) as an external instrument. The instrument must be relevant

$$E[Z_t u_{1t}] = \alpha \neq 0 \tag{6}$$

and it must be exogenous to the other structural shocks

$$E[Z_t \boldsymbol{u}_{2:nt}] = \boldsymbol{0}. \tag{7}$$

Here $\boldsymbol{u}_{2:nt} = (u_{2t} \dots u_{nt})'$ is the $(n-1)$ x 1 vector of all the other structural shocks. Under the two conditions above and the invertibility assumption, the structural impact vector $\boldsymbol{h}_1$ is identified up to a constant of proportionality (Stock and Watson, 2016; Baumeister and Hamilton, 2024). I resolve the scale ambiguity by defining an increase in the structural shock

u_{1t} as an event that raises the observed variable y_{1t} by δ units[4]

$$\widetilde{\boldsymbol{h}}_1 = (E[Z_t \boldsymbol{\epsilon}_t]/E[Z_t \epsilon_{1t}]) \times \delta \tag{8}$$

subject to $\boldsymbol{\Omega} = \boldsymbol{A^{-1}} \boldsymbol{D} (\boldsymbol{A^{-1}})'$. Thus, if $\widetilde{\boldsymbol{h}}_1$ is identified and the VAR is invertible, the structural shock u_{1t}, its variance d_{11}, and the forecast error variance decomposition are identified.

Dealing with data during COVID.— The inclusion of 2020 data poses substantial challenges when estimating the model. Following Lenza and Primiceri (2022), I conceptualize COVID as a regime in which the innovation variance of the VAR is scaled up, while the autoregressive dynamics are unchanged. Formally, I define a regime indicator $s_t \in \{1, 2\}$, where $s_t = 1$ denotes normal times and $s_t = 2$ denotes the COVID period. The autoregressive coefficients of the VAR are held constant across regimes, $\boldsymbol{\Pi}_1 = \boldsymbol{\Pi}_2 = \boldsymbol{\Pi}$, but there is a marked scale-up in the innovation variance, $\boldsymbol{\Omega}_2 = \kappa^2 \, \boldsymbol{\Omega}_1$, which generates large outliers. To implement this, I adopt a feasible Generalized Least Squares (GLS) iterative estimation procedure as suggested in Hamilton (Forthcoming), where the observations in the COVID sample are downweighted by the scaling factor κ^2, which is in turn estimated via Maximum Likelihood Estimation

$$\hat{\kappa}^2 = \frac{1}{T_2 \, n} \sum_{t=1}^{T} (\boldsymbol{y}_t - \boldsymbol{\Pi} \boldsymbol{x}_{t-1})' \, \boldsymbol{\Omega}_1^{-1} \, (\boldsymbol{y}_t - \boldsymbol{\Pi} \boldsymbol{x}_{t-1}) \, \mathbb{1}\{s_t = 2\}. \tag{9}$$

Here T_2 is the total number of observations in the COVID regime, and n is the number of endogenous variables considered in the model. The same procedure is applied to the two-stage least squares estimation of the structural impact vector $\widetilde{\boldsymbol{h}}_1$. In the empirical analysis, I define the COVID regime as the period from January to December 2020.

3.2 Empirical Specification

The baseline specification of the empirical model employed in the analysis is a monthly VAR which comprises two global variables, the Global Supply Chain Pressure Index (GSCPI), ordered first as it serves as the instrumented variable, and the West Texas Intermediate (WTI) oil price, along with three US variables: Consumer Price Index (CPI), industrial production, and unemployment rate. The sample period begins in January 1998, the earliest date for

[4] I anchor the global supply chain shock to the Global Supply Chain Pressure Index (GSCPI). Because the GSCPI is constructed by principal component analysis, its units are not intuitively interpretable. The instrument, by contrast, is measured in dollars per FEU container. Following Hamilton (Forthcoming), I use the slope coefficient from the first-stage regression of the GSCPI innovation on the surcharge to fix the scale of the shock in these dollar units.

which the Global Supply Chain Pressure Index is available, and ends in December 2024. However, the first-stage estimation is conducted over the shorter period from January 2014 to December 2024, as price announcement data from the three major shipping companies are unavailable before January 2014.

The variables in the model are transformed into logarithms multiplied by 100, with two exceptions: the Global Supply Chain Pressure Index, which is retained in its original units as standard deviations from the mean, and the unemployment rate, which is expressed as a percentage. The VAR includes 12 lags.

3.3 Aggregate Fluctuations

I now provide evidence of the propagation mechanism of a negative shock to the global supply chain identified via external instrument. Before turning to the results, I assess the strength of the instrument following Montiel Olea and Pflueger (2013). The heteroskedasticity-robust F-statistic is 12.40, exceeding conventional critical values, and I find no evidence of a weak instrument problem. I therefore proceed with standard inference. In all subsequent figures, the 68%, 80%, and 90% confidence bands are computed using 4,000 recursive moving-block bootstrap replications (Jentsch and Lunsford, 2019).

Figure 4 shows the structural Impulse Response Functions (IRFs) of the baseline model, normalized to a structural shock that increases the GSCPI by 0.41 units, corresponding to a $1066 price surcharge. This normalization corresponds to the surcharge recorded in December 2020. To contextualize this magnitude, the Freightos Baltic Index reports an average shipping cost for a FEU container of approximately $2,650 across major global routes in that month, implying that the $1066 surcharge represents roughly 40% of prevailing shipping costs.

After the initial increase, the GSCPI reverses back and returns to its steady state after approximately three years. US consumer prices exhibit a sluggish and persistent increase, consistent with a price friction mechanism, and the response is highly statistically significant. By contrast, oil prices show no statistically significant response at any horizon, indicating that the instrument is not picking up a commodity-price shock such as an oil shock. In Section 6, I address more comprehensively the concern that the instrument may capture oil supply or oil demand shocks rather than disruptions to the global supply chain. Turning to real economic effects, the shock induces an immediate decline in industrial production, which reaches its lowest point after two quarters before gradually recovering. This response is highly statistically significant and persistent, with industrial production returning to its steady state no earlier than three years post-shock. The unemployment rate rises immediately, peaks after

one year, and then gradually declines toward its steady state, remaining elevated for three additional years. At the peak, a supply-chain shock associated with a $1066 price surcharge raises consumer prices by almost 0.3%, while reducing industrial production by 0.76% and increasing unemployment by 0.14 percentage points.

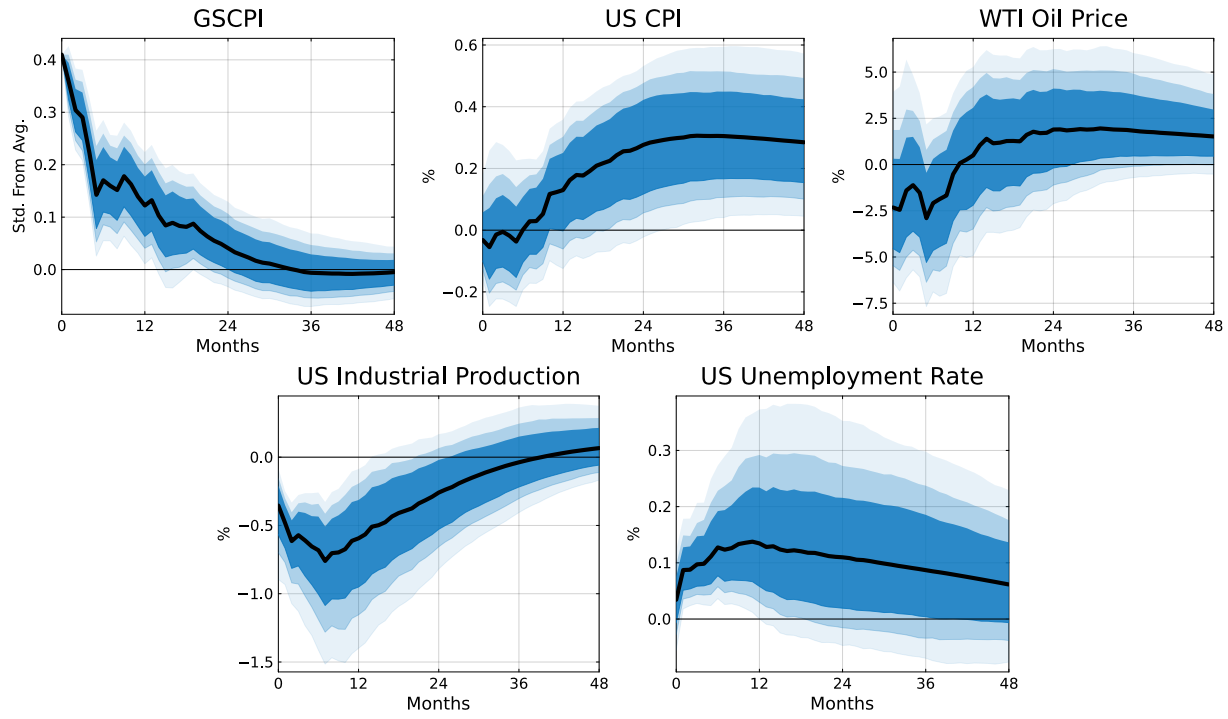


Figure 4: Structural Impulse Responses to a Global Supply Chain Shock.

Notes: Impulse responses to a global supply chain shock corresponding to a $1,066 price surcharge. The solid line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands, respectively. First stage sample from January 2014 to December 2024. First stage regression: F-statistic: 13.21, robust F-statistic: 12.40, R^2: 9.70%, Adjusted R^2: 8.97%.

Discussion.— At the aggregate level, a global supply chain shock functions as a cost-push shock. The most pronounced effect at impact is on production, which declines by 0.35% and takes nearly three years to return to its steady state. This persistence is consistent with the mechanisms emphasized by Alessandria et al. (2023), who show that temporary supply disruptions can generate protracted real effects. My findings also align with recent work quantifying the macroeconomic implications of supply chain disruptions. Bai et al. (Forthcoming) estimate relatively short-lived effects on real GDP and unemployment but persistent upward pressure on prices, patterns consistent with the price dynamics I obtain and with the evidence reported by Benigno et al. (2022) and Gordon and Clark (2023). These magnitudes are also broadly comparable to those in Känzig and Raghavan (2026), with the notable difference that the temporary nature of the identified shock generates a

transitory response of industrial production: production declines and then gradually returns to its steady state as pressure on the global supply chain eases.

3.4 Wider Effects

To gain a deeper understanding of the macroeconomic propagation of shocks to the global supply chain, I examine their effects across key sectors of the US economy. I analyze sectoral responses in industrial production across major subsectors, the response of the aggregate service sector, and the response of consumer prices disaggregated by CPI category. I estimate impulse response functions by incorporating each variable individually into the baseline model, maintaining the same sample period as in the baseline specification.

Industrial production.— Figure 5 presents the IRFs for eight key industrial sectors alongside the aggregate industrial production series. I examine the two primary components of industrial production by industry group, manufacturing and mining, as well as the major subcategories within manufacturing: durable and nondurable goods. Additionally, I focus on food and beverage and motor vehicles and parts, two sectors critical for business cycle analysis and among the largest components of nondurable and durable goods, respectively. I also analyze materials and final products, the two main components of industrial production by market group. Final products are related to immediate pass-through to consumer prices, while materials, as intermediate inputs, capture persistent second-round effects on firms' marginal costs.

Following a negative shock to the global supply chain, all industrial production variables decline immediately, reach their trough within the first few quarters, and then recover to pre-shock levels between two and three years later. All impulse response functions are highly statistically significant, except for mining. This exception is unsurprising: despite US reliance on imported rare earth materials and other mining products, mining production has a localized supply chain, resulting in a response to global supply chain shocks that is not statistically different from zero. Manufacturing displays a sharper peak contraction than the aggregate index (1.03% vs. 0.76%). At their respective troughs, durable goods fall more than nondurables (1.10% vs. 0.64%), with total manufacturing lying in between. Motor vehicles and parts exhibit the largest peak decline (3.52%), while food and beverages show the smallest (0.36%), with a quick return to steady state. Among market groups, the peak response of materials (0.94%) exceeds that of final products (0.64%).

To interpret the heterogeneity in response magnitudes, I employ the look-through exposure metric developed by Baldwin, Freeman and Theodorakopoulos (2023). Computed

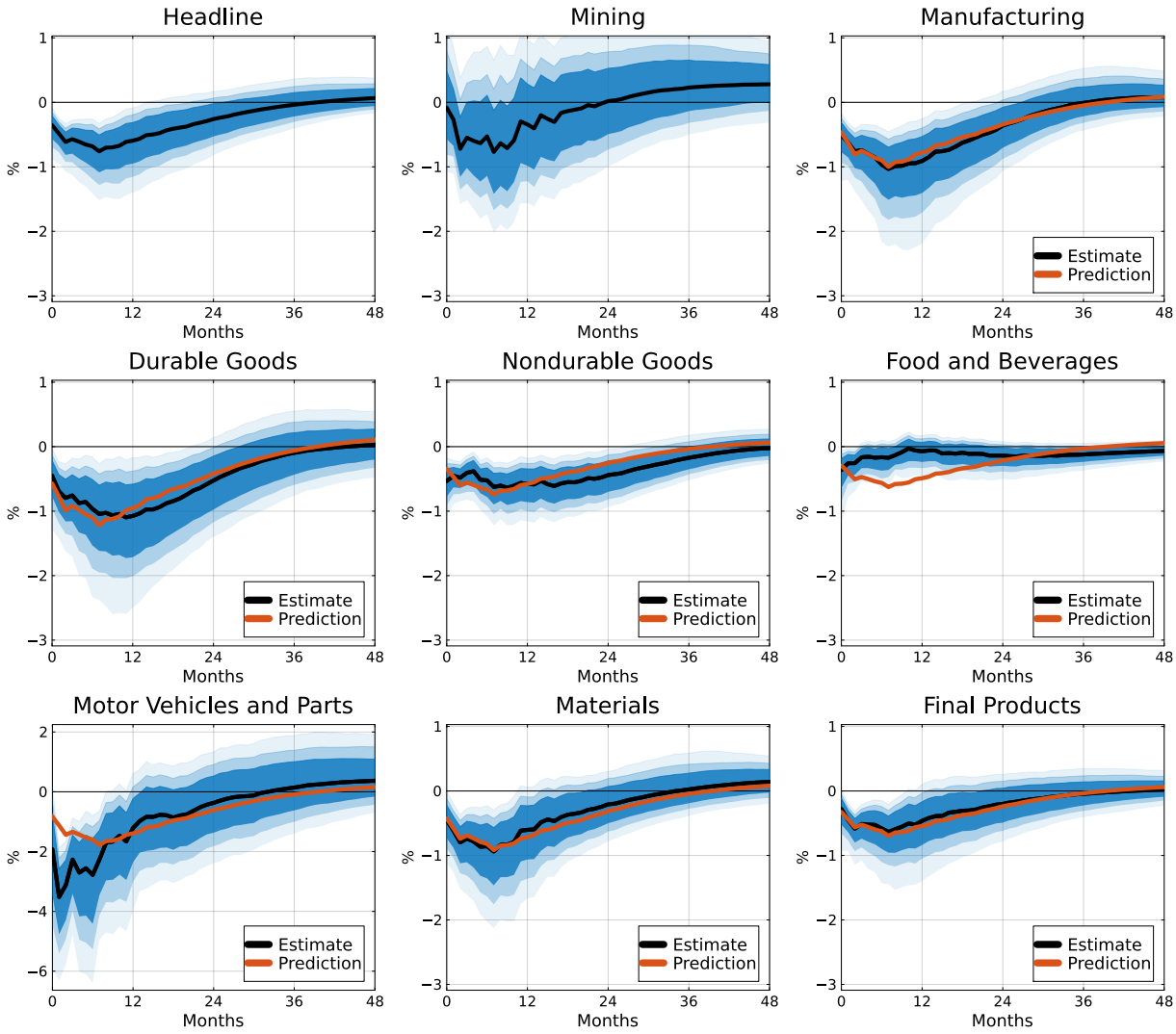


Figure 5: The Effects of a Global Supply Chain Shock on US Industrial Production.

Notes: Impulse responses to a global supply chain shock corresponding to a \$1066 price surcharge. The solid black line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands, respectively. The orange line is the predicted sectoral IRF, derived by rescaling the headline industrial production IRF with the ratio of sectoral to headline look-through exposure measures. All subplots share the same y-axis scale, except for Motor Vehicles and Parts.

from the OECD's 2021 ICIO tables via the Leontief inverse, it is the share of a sector's total manufactured intermediate inputs sourced from abroad, directly or indirectly, through its suppliers' suppliers. [5] Using this metric as a proxy for a sector's exposure to the global supply chain, I construct predicted impulse responses by rescaling the headline IRF proportionally

[5] Baldwin, Freeman and Theodorakopoulos (2023) compute look-through exposure for 17 NAICS 31–33 manufacturing subsectors. I calculate exposure for broader manufacturing aggregates and the two market group series, using weights from the Industrial Production and Capacity Utilization (G.17) statistical release by the Board of Governors of the Federal Reserve System.

to each sector's relative exposure:

$$\widetilde{\text{IRF}}_i = \widehat{\text{IRF}}_h \times \frac{\text{LT}_i}{\text{LT}_h}. \tag{10}$$

Here, $\widehat{\text{IRF}}_h$ is the impulse response function of the headline series, and LT_i and LT_h represent the look-through exposure for sector i and the headline, respectively.

These predicted responses, shown as orange lines in Figure 5, closely match the estimated sectoral IRFs, indicating that relative look-through exposure alone accounts for most of the cross-sectoral dynamics. Manufacturing, with a look-through exposure of 12.3%, is more exposed than the aggregate index, explaining its larger response. Durable goods are more exposed (15%) than nondurables (9.1%), and the total manufacturing response falls between the two, reflecting its weighted average composition. Motor vehicles and parts, with the highest exposure (21.9%), exhibit the most severe contraction, while food and beverages, with the lowest exposure (7.7%), experience the mildest. For food and beverages, the predicted IRF captures the response well on impact but overstates its persistence. This muted and short-lived response reflects the sector's limited reliance on foreign inputs and the prevalent use of capital goods that operate over multiple production cycles. Lastly, the different exposure of materials (11.3%) and final products (8.6%) explains the difference in their responses.

Services.— To assess the broader macroeconomic reach of global supply chain shocks, I examine their impact on the non-tradable sector, specifically services. Figure 6 presents the impulse responses of the ISM Services Activity Index, ISM Services Employment Index, and ISM Manufacturing Production Index to a negative shock corresponding to a $1066 price surcharge. Both services indicators exhibit no statistically significant response on impact, consistent with services not being directly exposed to global supply chain disruptions. However, approximately two quarters later, they decline significantly before returning to pre-shock levels within a year.

Including the ISM Manufacturing Production Index allows a direct comparison across tradable and non-tradable sectors, since all three are survey-based diffusion indices measured on comparable scales. Manufacturing production drops sharply on impact, reflecting the tradable sector's direct exposure, whereas the services response is both lagged and more muted. Because the ISM indices enter the VAR in logs (times 100), these responses should be read as percentage deviations from their steady-state level: a decline signals a broad-based weakening of activity.

Consumer prices.— The broad-based output declines documented above are mirrored by

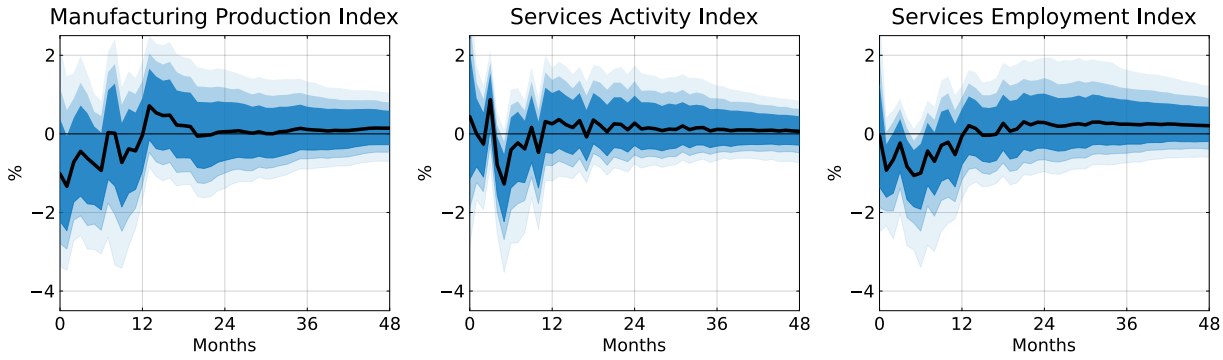


Figure 6: The Effects of Global Supply Chain Shock on Manufacturing and Non-Tradables.

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The solid black line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands, respectively. All subplots share the same y-axis scale.

equally pervasive inflationary pressures, with sectors more exposed to the global supply chain tending to exhibit stronger and more persistent pass-through to consumer prices. Figure 7 shows the responses of different components of CPI, including core, services, durable and nondurable goods, food and beverages, new vehicles, housing, and medical care, together with the headline response from the baseline model. A consistent pattern emerges across all components: each category exhibits a sluggish and persistent increase, with many remaining elevated even four years after the shock. This broad-based inflationary response reflects a gradual transmission from production disruptions to consumer prices, consistent with the presence of nominal rigidities.

The core CPI response is nearly identical to the headline CPI, reflecting the broad-based nature of the inflationary impact. This close alignment is consistent with the widespread decline observed across core production sectors.

The services CPI component exhibits a relatively mild yet persistent increase in response to a global supply chain shock. This inflationary pattern aligns with the observed declines in service sector activity and employment documented in Figure 6. Given that services are only indirectly exposed to global supply chain disruptions, the associated contraction in output is more limited. Accordingly, the pass-through to consumer prices is less pronounced, though still persistent.

Durable and nondurable goods exhibit markedly different inflationary responses, mirroring their respective production dynamics shown in Figure 5. Durables, which are more exposed to global supply chains, experience a stronger and more persistent price increase. This parallels the sharper and more prolonged contraction in durable manufacturing, particularly in sectors such as motor vehicles and parts. Nondurables, with lower exposure, exhibit

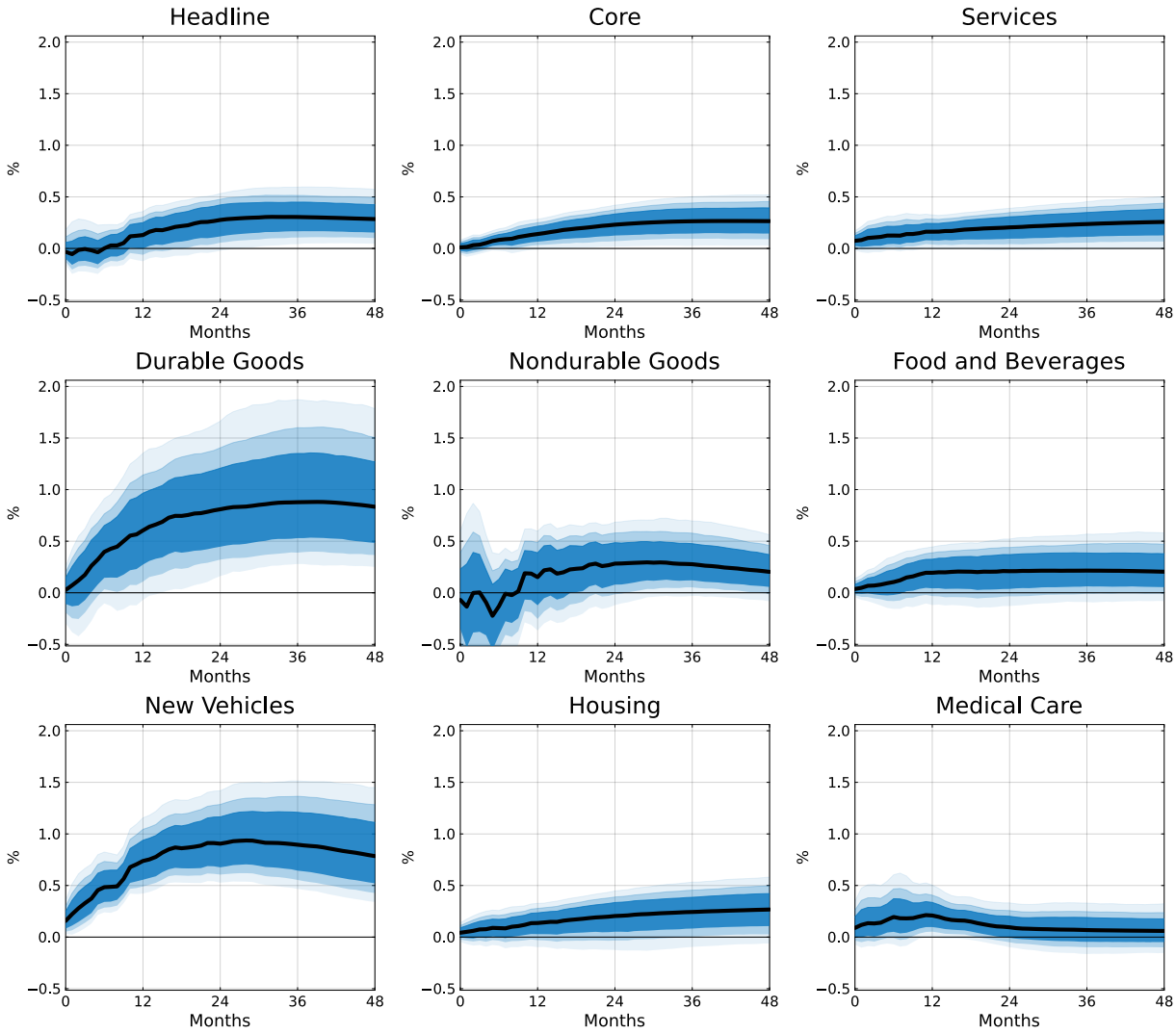


Figure 7: The Effects of Global Supply Chain Shock on US Consumer Prices.

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The solid black line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands, respectively. All subplots share the same y-axis scale.

a more modest price increase.

The divergence is even more pronounced when comparing food and beverages to new vehicles. The food and beverage category, which shows the smallest production response due to its localized supply chain structure, also displays the mildest CPI increase. By contrast, new vehicles, embedded in a highly globalized and input-intensive supply chain, exhibit the largest and most persistent CPI response, closely tracking the steep and extended production decline in motor vehicles and parts.

Lastly, I include housing and medical care to illustrate how sectors that appear largely

insulated from international trade can still be affected by global supply chain disruptions. In addition, housing is the largest component of the CPI basket, making its inflationary dynamics particularly interesting. Both categories show persistent price increases following the shock, underscoring the economy-wide reach of supply chain disruptions, even in sectors not traditionally viewed as integrated in the global supply chain.

Discussion.— Sectoral responses to a global supply chain shock exhibit similar qualitative dynamics but vary significantly in magnitude. Among production sectors, this heterogeneity is strongly correlated with each sector's degree of exposure to the global supply chain. Predicted impulse response functions, obtained by rescaling the IRF of aggregate industrial production by relative look-through exposures, closely match the estimated sectoral dynamics. This suggests that look-through exposure serves as a sufficient statistic for explaining cross-sectoral variation. Sectors experiencing the most pronounced output contractions, such as durable manufacturing, and in particular motor vehicles, also exhibit the strongest pass-through to consumer prices. While all CPI components respond persistently to the shock, durable goods and new vehicles stand out in both the size and persistence of their inflationary dynamics.

Beyond the tradable goods sectors, I find that global supply chain shocks spill over into non-tradable services. Although the responses of service activity and employment are more modest and transitory, they confirm the broader macroeconomic reach of these shocks. Services CPI also rises persistently, reflecting higher operational costs that are gradually passed on to consumers despite the sector's lower direct exposure to the global supply chain.

My sectoral analysis highlights why shocks to the global supply chain differ fundamentally from other cost-push shocks, such as oil supply shocks, among the most studied in the literature. Oil shocks generate highly heterogeneous sectoral production responses, driven by factors like energy intensity in production, energy intensity of the goods produced, and demand substitutability (Davis and Haltiwanger, 2001; Hamilton, 2009; Herrera, Lagalo and Wada, 2011). Most industrial sectors show limited output responses to oil shocks, with price effects varying: some consumer price categories exhibit persistent increases, others are transitory, and some remain unaffected (Tenreyro, 2022; De Santis, 2024). In contrast, my results show that global supply chain disruptions lead to widespread output losses and price increases across all sectors. These broad-based sectoral effects reflect the breadth of my instrument, which captures disruptions to containerized intermediate and final goods used across the economy. By contrast, a dry-bulk measure such as that of Känzig and Raghavan (2026) produces more heterogeneous sectoral responses, since exposure to bulk shipping varies with each sector's reliance on the underlying commodities.

4 Global Supply Chain Shocks and The COVID Era

The causes of the pandemic-era inflation have sparked intense debate in the literature, with competing explanations emphasizing either demand- or supply-side factors. Shocks to the global supply chain have been increasingly acknowledged as a contributing factor, with existing assessments based on sign-restricted or narrative sign-restricted SVARs.

In this section, I complement this evidence by providing a comprehensive analysis of the cumulative historical contribution of global supply chain shocks to the rise of consumer prices and the recovery of industrial production from January 2020 to December 2024, leveraging the instrument and structural framework developed in Sections 2 and 3. In particular, I will answer two key questions: how significantly global supply chain shocks contributed to post-pandemic inflation and how these shocks influenced the recovery of economic activity.

4.1 Historical Decomposition

Given the econometric framework outlined in Section 3, if the structural impact vector $\widetilde{\boldsymbol{h}}_1$ is identified, Hamilton (Forthcoming) shows that the value of the first structural shock at date t, u_{1t}, can be recovered from the vector of the reduced-form innovations $\boldsymbol{\epsilon}_t$ as follows

$$u_{1t} = \left(\widetilde{\boldsymbol{h}}_1' \boldsymbol{\Omega}^{-1} / \, \widetilde{\boldsymbol{h}}_1' \boldsymbol{\Omega}^{-1} \widetilde{\boldsymbol{h}}_1 \right) \boldsymbol{\epsilon}_t \, . \tag{11}$$

Using the identified structural shock, I can estimate the contribution of the series of global supply chain shocks to the historical values of the variables included in the VAR model. The structural moving average representation from equation (5) implies:

$$\boldsymbol{y}_t(u_1) = \sum_{h=0}^{t-1} \boldsymbol{\Psi}_h \widetilde{\boldsymbol{h}}_1 u_{1t-h}, \tag{12}$$

where t indexes the observations from the start of the decomposition window. Here, $\boldsymbol{y}_t(u_1)$ is referred to as historical decomposition, obtained as the sum of the realized global supply chain shocks u_{1t-h} weighted by the corresponding structural impulse response coefficients $\boldsymbol{\Psi}_h \widetilde{\boldsymbol{h}}_1$.

4.2 Post-Pandemic Inflation

Figure 8 shows the cumulative historical contribution of global supply chain shocks to the consumer price level from January 2020 to December 2024, measured in percentage deviations from the January 2020 level. The figure reveals that global supply chain shocks were key drivers of post-pandemic inflation, particularly during the first half of the sample period. In

2021, global supply chain shocks accounted for an average of 45% of the surge in consumer prices, reflecting the scale of the unprecedented disruptions worldwide.

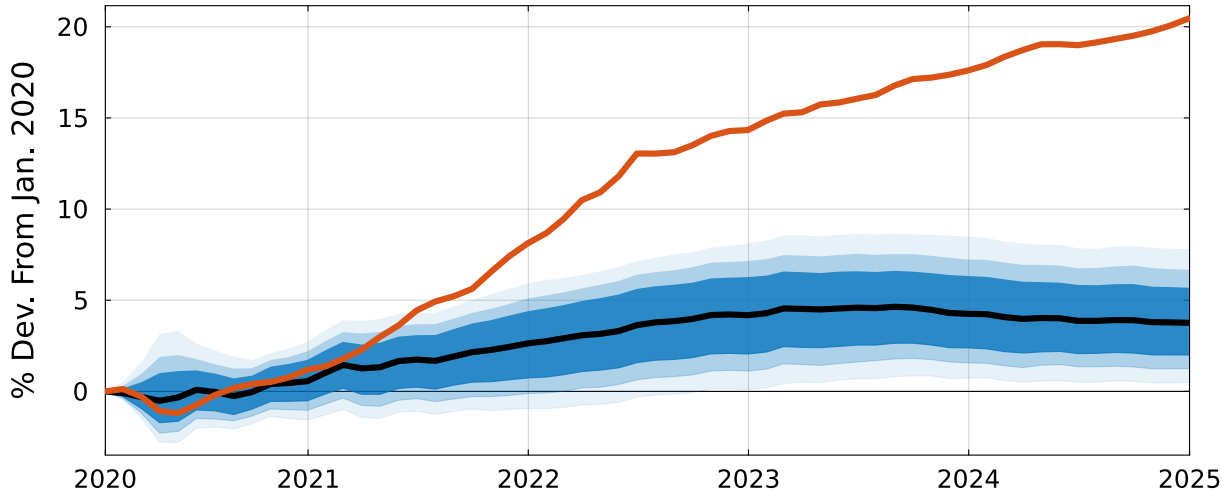


Figure 8: Historical Decomposition of Consumer Prices

Notes: The figure illustrates the historical contribution of global supply chain shocks to the US CPI, expressed as a percentage deviation from the pre-pandemic level (January 2020). The solid orange line is the actual series, the solid black line is the point estimates, and the shaded areas are 68, 80 and 90 percent confidence bands, respectively.

Beginning in the second half of 2020, severe disruptions at major hubs, such as the ports of Los Angeles and Long Beach (handling a combined 8.6 million FEUs annually), emerged alongside disruptions in the United Kingdom, Australia, and New Zealand. The delayed pass-through of these shocks triggered significant price increases in early 2021. The impact intensified through 2021 and 2022, exacerbated by global supply chain shocks associated with events such as the partial closure of China's Yantian terminal in Shenzhen Port (7 million FEUs annually, ranking 4th globally) from May to June 2021, and the shutdown of the Meishan terminal at Ningbo-Zhoushan port (16.3 million FEUs annually, 3rd globally) in August 2021, disrupting 25% of its container cargo capacity. The March 2021 Suez Canal obstruction, blocking a chokepoint responsible for 12% of global trade, further strained the system. In early 2022, stringent lockdown measures in Shenzhen and Shanghai, which impacted Yantian and Shanghai ports (the world's busiest, handling 21.8 million FEUs annually), and the disruption of the Black Sea trade following the Russian invasion of Ukraine, led to the peak contribution to consumer prices observed in early 2023. Although the impact began to subside thereafter, new disruptions persisted into 2024, notably the January suspension of Red Sea transit, which forced rerouting via the Cape of Good Hope.

My findings confirm the primary role of global supply chain shocks in shaping the early

post-pandemic inflation dynamics documented by Bai et al. (Forthcoming). They are also consistent with the conclusions of Comin, Johnson and Jones (2023) and Caldara, Iacoviello and Yu (2025), who emphasize that shortage shocks and supply constraints, closely related to the disruptions captured by my proposed instrument, were key drivers of inflation during 2021. Similarly, Shapiro (2024) highlights the dominant contribution of supply-side factors in the early phase of the inflation surge, while Rubbo (2024) stresses the importance of aggregate supply shocks. My findings are also in line with Hobijn and Sahin (2022), whose decomposition of labor market aggregates suggests that the pandemic-induced labor supply shock contributed less to the early stage of post-pandemic inflation.

On the other hand, Di Giovanni et al. (2022) attribute the majority of the price surge in 2020-2021 to demand-side forces, a conclusion reaffirmed in their subsequent work (Di Giovanni et al., 2023*b*,*a*). The discrepancy most likely reflects differences in methodology: their model-based quantitative exercise, building on Baqaee and Farhi (2022), relies on feeding the model with demand and supply shocks constructed from calibration exercises or linear detrending methods.

Thus, while global supply chain shocks accounted for a large share of the 2021 inflation surge, the remaining variation could be largely attributable to demand factors and the substantial fiscal deficit, which makes my results consistent with the interpretation of Jordà and Nechio (2023), Garcia Revelo, Levieuge and Sahuc (Forthcoming), Faria-e Castro (2024), Giannone and Primiceri (2024), Hazell and Hobler (2024), and Bergholt et al. (Forthcoming). In the subsequent years, the contribution of global supply chain shocks to inflation moderated while remaining significant, at approximately 29% in 2022, 28% in 2023 and 21% in 2024. This also aligns with findings from Goda and Soltas (2022), and Lee, Park and Shin (2023), who demonstrate that adverse shocks to productive capacity sustained elevated inflation levels from 2022.

4.3 Recovery of Industrial Production

Figure 9 shows the cumulative historical contribution of global supply chain shocks to industrial production for the period January 2020 - December 2024, measured in percentage deviations from the production level observed in January 2020. The figure highlights the substantial downward pressure these shocks exerted on the economic recovery. By November 2021, industrial production had returned to its pre-COVID level, but in a counterfactual scenario with only global supply chain shocks, recovery would have been delayed until at least May 2023.

From January 2020, following reports of the COVID-19 outbreak in China, major ship-

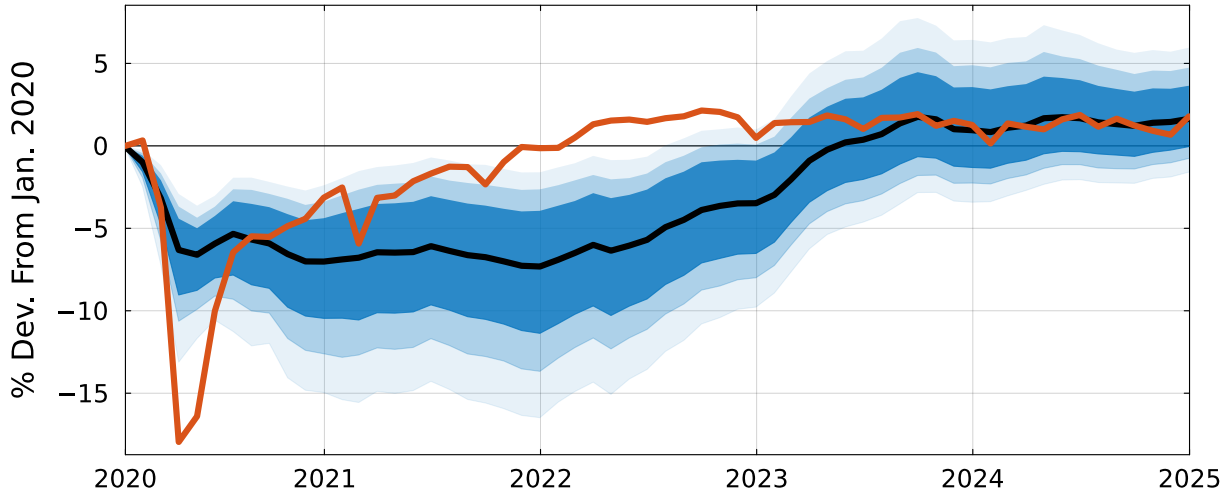


Figure 9: Historical Decomposition of Industrial Production

Notes: The figures show the historical contribution of global supply chain shocks to the US Industrial Production in percentage deviation from the pre-pandemic level (January 2020). The solid orange line is the actual series, the solid black line is the point estimates, and the shaded areas are 68, 80 and 90 percent confidence bands, respectively.

ping companies implemented extraordinary blank sailing omissions on key trade routes, notably on the Trans-Pacific and East-West corridors. By February 2020, severe disruptions in Chinese ports, as reported by MSC, led to critical shortages of refrigerated container plugs and prevented vessels from discharging at designated ports. These disruptions caused significant delays in the shipment of intermediate goods and sharply increased production costs.

In May 2020, global supply chain shocks accounted for nearly 40 percent of the total decline of industrial production, equivalent to a contraction of about 6.6 percent. This estimate is in line with the quantitative exercises of Brinca, Duarte and Faria-e Castro (2021) and Baqaee and Farhi (2022), who emphasize the substantial drag exerted by supply-side forces on economic activity during the early phase of COVID-19. While industrial production began to recover in the second half of 2020, the downward pressures from supply chain shocks intensified. Their contribution to the production shortfall peaked at -7.25% in December 2021, in line with the findings of Alessandria et al. (2023). Their effect gradually declined thereafter, with the negative contribution dissipating by May 2023.

4.4 Summary

The historical decomposition analysis reveals that global supply chain shocks played a more prominent role in the early phase of post-pandemic inflation than previously recognized. My

findings confirm the central importance of these shocks in shaping inflation dynamics in 2021. However, I do not dismiss the role of demand-side forces: large-scale fiscal stimulus and pent-up demand also contributed meaningfully to the inflationary surge. In particular, beginning in the second half of 2021, my results suggest that other shocks increasingly became the dominant drivers of the price level, consistent with the timing of the March 2021 American Rescue Plan.

Finally, this analysis contributes a novel insight to the ongoing policy debate: while much of the discussion on global supply chain shocks has centered on their inflationary consequences (Tenreyro, 2021; Lane, 2022; Guerrieri et al., 2023), my findings show that these shocks also have long-lasting contractionary effects on real activity. This persistence highlights the importance of recognizing these shocks not only as a source of persistent inflation but also as a meaningful driver of output fluctuations over the business cycle.

5 Quantitative Importance

Forecast Error Variance Decomposition.— The preceding analysis has shown that global supply chain shocks are not only a source of persistent inflation but also a meaningful driver of output fluctuations over the business cycle. Building on this insight, I now turn to a formal quantification of their contribution to macroeconomic volatility. Specifically, I assess the average importance of global supply chain shocks in explaining forecast error variance across the aggregate variables in the baseline model. This allows me to evaluate how much of the business cycle is systematically accounted for by these shocks over different time horizons.

Under the identification assumptions of Section 3, if the structural impact vector $\widetilde{\boldsymbol{h}}_1$ is identified, the variance of the shock is also identified as

$$d_{11} = \left(\widetilde{\boldsymbol{h}}_1' \boldsymbol{\Omega}^{-1} \widetilde{\boldsymbol{h}}_1\right)^{-1} \tag{13}$$

where $\boldsymbol{\Omega}$ is the reduced-form variance-covariance matrix. Consequently, I compute the contribution of the global supply chain shock to the forecast error variance-covariance matrix of $\boldsymbol{y}_t$ at horizon h as

$$\sum_{s=0}^{h} \boldsymbol{\Psi}_s \widetilde{\boldsymbol{h}}_1 d_{11} \widetilde{\boldsymbol{h}}_1' \boldsymbol{\Psi}_s'. \tag{14}$$

The ratio of the diagonal elements of (14) to those of the overall forecast error variance-covariance matrix at horizon h, $\sum_{s=0}^{h} \boldsymbol{\Psi}_s \boldsymbol{\Omega} \boldsymbol{\Psi}_s'$, provides the percentage of the variance of $\boldsymbol{y}_t$ at horizon h accounted for by the first structural shock.

Table 1: Forecast Error Variance Decomposition

	US CPI	WTI Oil Price	US Ind. Prod.	US Unemp.	GSCPI
Horizon					
0	0.01 [0.00, 0.20]	0.04 [0.00, 0.24]	0.16 [0.03, 0.54]	0.01 [0.00, 0.18]	0.78 [0.49, 0.99]
12	0.02 [0.00, 0.17]	0.02 [0.00, 0.16]	0.25 [0.08, 0.62]	0.13 [0.03, 0.47]	0.74 [0.49, 0.96]
24	0.11 [0.03, 0.37]	0.03 [0.00, 0.19]	0.24 [0.08, 0.61]	0.11 [0.03, 0.48]	0.66 [0.45, 0.85]
36	0.21 [0.10, 0.51]	0.05 [0.00, 0.21]	0.24 [0.08, 0.60]	0.11 [0.03, 0.45]	0.60 [0.40, 0.79]
48	0.26 [0.14, 0.58]	0.06 [0.00, 0.23]	0.24 [0.08, 0.59]	0.11 [0.02, 0.43]	0.59 [0.40, 0.78]
∞	0.30 [0.15, 0.70]	0.08 [0.00, 0.26]	0.24 [0.09, 0.56]	0.10 [0.01, 0.38]	0.58 [0.40, 0.78]

Note: The table shows the proportion of the forecast error variance of the variables in the baseline model explained by global supply chain shocks at horizons 0, 12, 24, 36, 48 months and asymptotically. The 90 percent confidence intervals are displayed in brackets.

Discussion.— Table 1 presents the results. The global supply chain shock accounts for the vast majority of fluctuations in the global supply chain pressure index, explaining almost 80% of its variance on impact and nearly 60% in the long run. In contrast, the shock contributes minimally to oil price variability, with almost none of the short-term variation attributed to global supply chain disruptions, confirming that oil prices are largely disconnected from such shocks. The shock's contribution to consumer price fluctuations is small in the first year but grows over the long term, accounting for 30% of CPI variability in the long run. Turning to real variables, it explains a substantial portion of industrial production variability, increasing from 16% on impact to 24% in the long run. The shock's contribution is modest for unemployment on impact, but rises to about 10% after one year and in the long run. My estimates are in line with those of Bai et al. (Forthcoming) on the nominal side, while attributing to these shocks a more prominent role on the real side of the economy.

Taken together with the impulse response and historical decomposition analyses, these results show that global supply chain shocks are key drivers of business cycle fluctuations. One of the motivations for developing an external instrument was the difficulty of conducting causal inference with aggregate indices of supply chain pressure, which are themselves endogenous to fluctuations in global demand and production activity. A simple exercise in

Online Appendix C shows that this endogeneity concern is first order. Using a recursive ordering identification strategy on the GSCPI, global supply chain shocks have essentially no real effects: production and unemployment do not respond, the shocks contribute almost nothing to the pandemic-era production shortfall, and they do not explain business-cycle fluctuations in real activity.

6 Sensitivity Analysis

This section provides additional evidence on the validity of the identification strategy and the robustness of empirical results.

6.1 Predictability of the Instrument

One test of whether the reported surcharges could be an endogenous response to underlying economic conditions is to see if they can be predicted by other economic indicators or by their own lagged values.

I conducted a battery of Granger causality tests on a wide range of variables, reported in Online Appendix A.1 Table A.1, including real activity indicators (industrial production, employment), prices (consumer prices, oil prices), interest rates and financial conditions, and widely used geopolitical-risk and macroeconomic uncertainty indices. The null hypothesis that lagged values of each control do not predict the proxy is not rejected variable-by-variable, and the corresponding joint test also fails to reject.

I rule out serial correlation in the FEU price surcharge series through a series of Ljung–Box tests, evaluated at multiple lags (see Online Appendix A.2 Table A.3). At each horizon, the joint null that the autocorrelations up to that lag are zero is not rejected. The FEU price surcharge series satisfies both conditions required of a valid external instrument.

6.2 Robustness with Respect to Category of Events

A natural concern is that certain categories of disruptions may not be fully exogenous, labor strikes could reflect endogenous responses to economic conditions; war-related events may affect the economy through channels beyond shipping, or may carry distinct macroeconomic implications that drive the aggregate results. A natural robustness check is to assess whether any single category drives the results.

Figure 10 reports the aggregate IRFs obtained by sequentially excluding one category of disruptions from the instrument. The findings remain unchanged relative to the baseline reported in orange, ruling out that any individual class of disruptions drives the results.

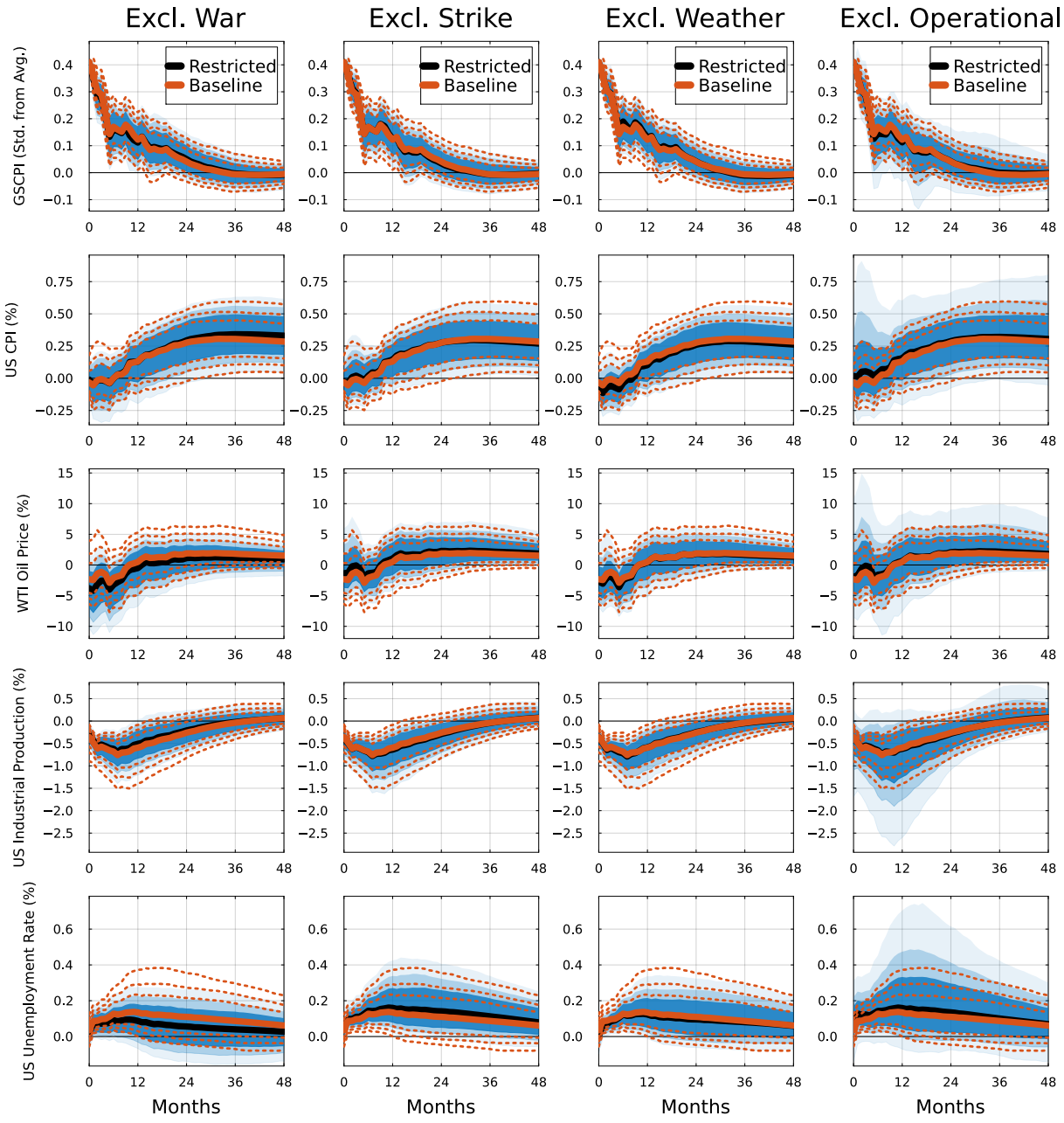


Figure 10: Robustness to Excluding Categories of Supply Chain Disruptions

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. Each column shows estimates excluding one category of surcharges from the instrument. The heteroskedasticity-robust effective first-stage F-statistic equals 12.40 for the baseline instrument, 17.87 when war events are excluded, 13.06 when strike events are excluded, 11.94 when climate events are excluded, and 3.08 when operational events are excluded. The orange lines represent the baseline specification, while the black lines represent the restricted estimates excluding the specified category. The solid lines represent the point estimates, and the shaded areas (and dotted lines) denote 68, 80, and 90 percent confidence bands.

The heteroskedasticity-robust F-statistic of the first-stage regression is above the conventional weak-instrument threshold when excluding war, strike, and weather subsets, and is lower only for the no-operational variant (see Online Appendix A.3 Table A.4). The exclusion of operational disruptions is the most demanding cut. Events in this category tend to involve some of the most crucial ports in the global system, Los Angeles, Long Beach, Ningbo, Nhava Sheva, Colombo, and the Suez Canal, among others, and their removal reduces the non-zero observations in the identification sample by 30%. The mechanical loss of first-stage strength when these specific events are removed is therefore unsurprising and does not cast doubt on the robustness of the main results: their exclusion does not alter the shape of the impulse responses, only the precision of the estimates. The dynamic causal effects of the global supply chain shock are not an artefact of any single category of disruption.

6.3 Other Indications of Exogeneity

A second concern is that the instrument might capture variation that can be attributed to demand conditions or other structural shocks. Several pieces of evidence weigh against this.

The narrative analysis, detailed in the Online Appendix, is designed precisely to prevent such contamination. Every candidate surcharge is cross-validated against independent external sources, and any event for which the stated cause could be directly linked to other structural shocks, whether on the demand or supply side, is excluded from the proxy. Moreover, the majority of the retained events are confined to individual ports rather than major trade routes or entire countries, a narrow geographic and operational footprint that makes it implausible they coincide with global structural shocks that drive macroeconomic aggregates. The sectoral results in Section 3 confirm that this screening is effective: every subcategory of real activity contracts and every subcategory of consumer prices rises in response to the shock. This pattern is the hallmark of a supply shock.

I also address the concern that the size of a given surcharge could, in principle, reflect contemporaneous demand pressures even when the disruption event is supply-side. I construct an additional instrument as a $\{+1, 0, -1\}$ event indicator that records the direction of each price surcharge announcement while abstracting from its magnitude. Figure 11 reveals the robustness of the stagflationary effects of global supply chain shocks, although confidence bands widen, reflecting the loss of information embedded in the surcharge magnitudes.

I corroborate this evidence with a series of Granger-causality tests, reported in Table 2. For China, the Euro Area, Japan, S. Korea, Taiwan, the United Kingdom, and the United States, I test the predictive power of the two demand subcomponents of the manufacturing PMI, new orders and quantities purchased, capturing customer demand for firms' products

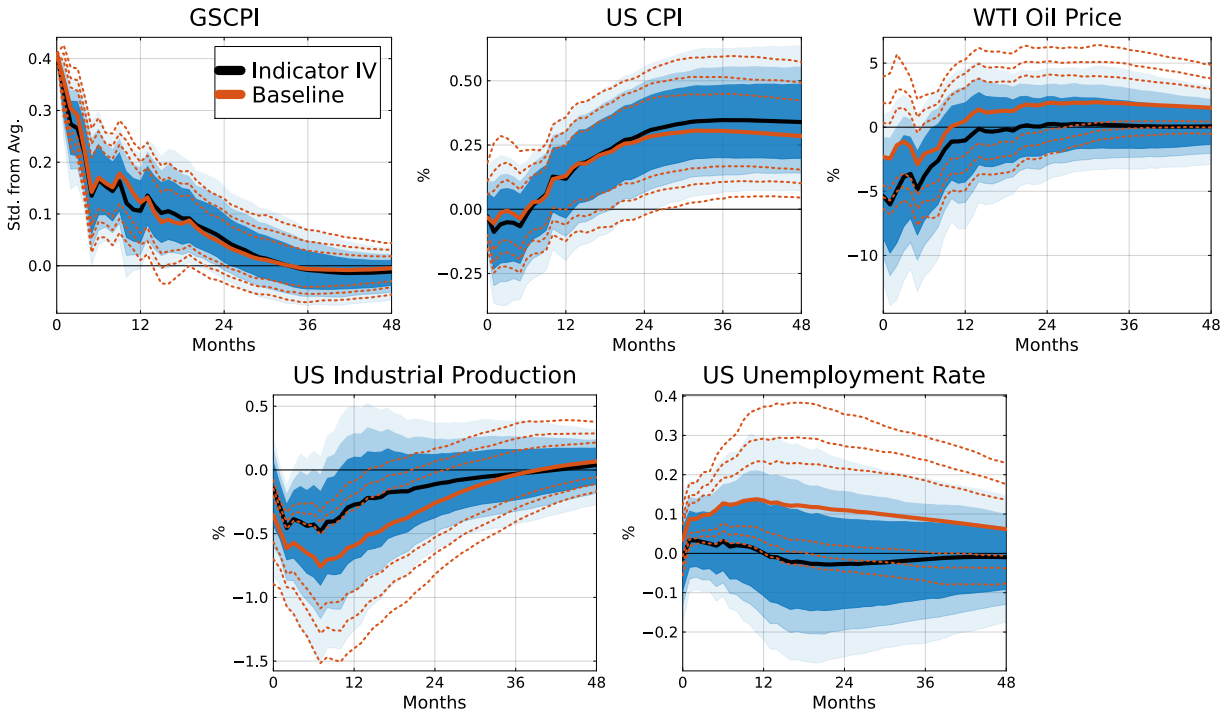


Figure 11: Robustness to a $\{+1, 0, -1\}$ Event-Indicator Instrument

Notes: Impulse responses to a global supply chain shock corresponding to a \$1066 price surcharge. The orange lines represent the baseline specification; the black lines represent the estimates obtained when the instrument is rebuilt as a $\{+1, 0, -1\}$ indicator of the occurrence of a disruption. The solid lines are point estimates, and the shaded areas (and dotted lines) denote 68, 80, and 90 percent confidence bands.

and firms' demand for intermediate inputs; these are the series Benigno et al. (2022) use to purge demand factors from the inputs of the GSCPI. I further test the demand-influenced supply-chain conditions, namely supplier delivery times, purchased stocks, input prices, and backlogs, together with the main shipping-cost indices: the Baltic Dry Index, the HARPEX container-charter index, and BLS air-freight rates. None of these variables predicts the instrument. Were surcharge magnitudes systematically related to demand conditions, one would expect these variables to help predict it.

These findings align with how these carriers price. As oligopolists, they reprice to demand continuously — through the general rate increases and peak-season surcharges documented in Section 2 — so demand pressure is passed through in real time, not withheld and released under cover of a supply-side disruption. The disruption surcharges I isolate therefore sit separately, on top of this routine repricing.

Lastly, I study the correlation between the FEU price surcharge series and a wide catalogue of structural shocks from the literature, including oil supply and oil demand shocks,

monetary policy shocks, financial shocks, and macroeconomic uncertainty shocks (see Online Appendix A.1 Table A.2). None of the correlations is statistically significant. As a further check, motivated by the concern that events such as those at the Strait of Hormuz might reflect oil-market shocks, I augment the baseline VAR with an identified oil shock—either the oil supply news shock of Känzig (2021) or the oil supply shock of Baumeister and Hamilton (2019)—one at a time, and find the impulse responses essentially unchanged (Online Appendix Figure A.1)

Table 2: Granger-causality tests (p-values) for shipping- and supply-chain-related variables.

Panel A: PMI Subcomponents							
	China	Euro Area	Japan	S. Korea	Taiwan	UK	US
Delivery Times	0.66	0.72	0.86	0.16	0.78	0.79	0.44
Purchased Stocks	0.16	0.57	0.37	0.27	0.49	0.79	0.37
Input Prices	0.93	0.68	0.37	0.55	0.57	0.63	0.34
New Orders	0.41	0.10	0.28	0.18	0.50	0.32	0.15
Backlogs	0.90	0.15	0.93	0.47	0.56	0.44	0.15
Quantity Purchased	0.33	0.20	0.45	0.19	0.35	0.59	0.22
Panel B: Shipping-cost							
Baltic Dry Index	0.58						
HARPEX	0.18						
Air Freight Asia In	0.10						
Air Freight Asia Out	0.87						
Air Freight Europe In	0.55						
Air Freight Europe Out	0.62						
Joint	0.79						

Note: Granger-causality tests of the FEU price surcharge series on shipping- and supply-chain-related variables. Every individual test regresses the instrument on a constant, its own 2 lags, and 2 lags of the series, on the common sample 2014M10–2023M1. The *Joint* row tests all series simultaneously with 1 lag each and 2 instrument lags. Source: Bloomberg. * p<0.05, ** p<0.01, *** p<0.001.

6.4 Exclusion Restriction

Another concern is that disruptions at US ports can, in principle, violate the exclusion restriction through a direct domestic channel: a strike or closure at a US port disrupts domestic logistics, inventories, and employment directly, not only through global supply chain conditions. Reassuringly, only five of the 54 disruptions in the instrument occur at US ports. To assess whether they drive the results, I reconstruct the instrument excluding these five events; Figure 12 compares the resulting impulse responses with the baseline.

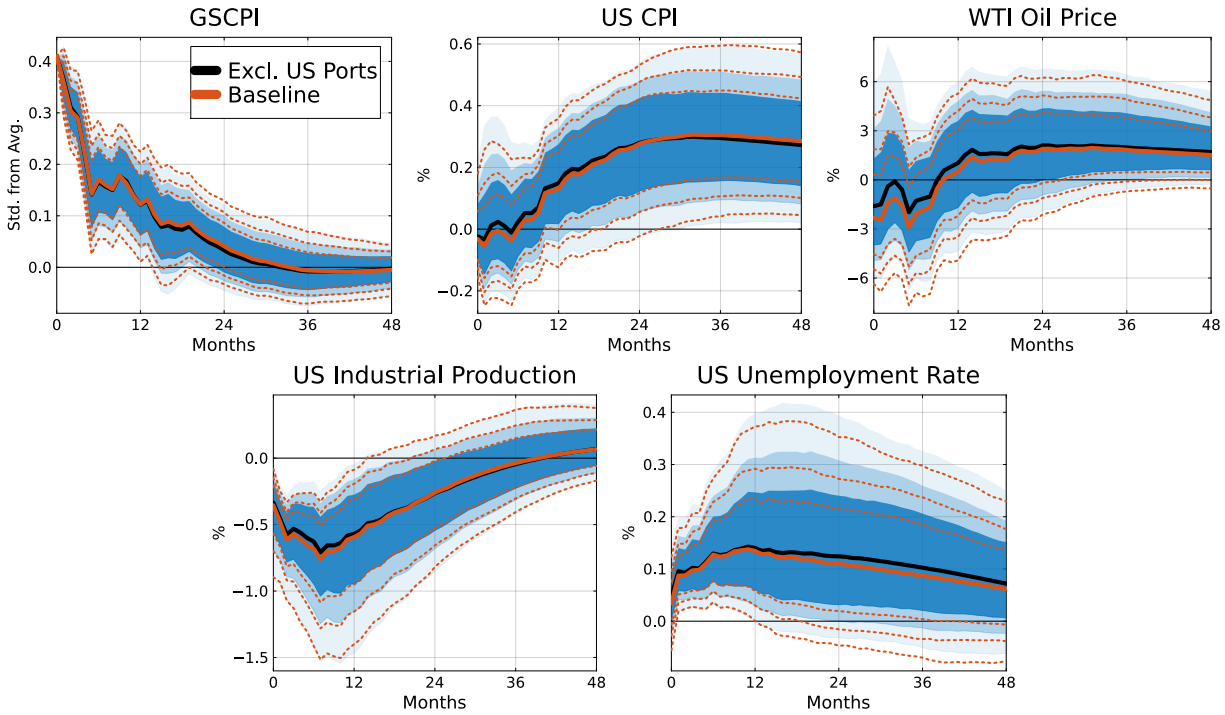


Figure 12: Robustness to Excluding US-Port Disruptions

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The orange lines represent the baseline specification, while the black lines represent the estimates obtained when the five disruptions at US ports are excluded from the instrument. The heteroskedasticity-robust first-stage F-statistic equals 10.85 for this restricted instrument, compared with 12.40 for the baseline. The solid lines represent the point estimates, and the shaded areas (and dotted lines) denote 68, 80, and 90 percent confidence bands.

The two sets of responses are essentially indistinguishable at every horizon, and the first stage remains strong. This evidence rules out the presence of a direct domestic channel.

6.5 Is COVID Driving the Results?

The COVID-19 period raises two distinct concerns: contamination of the VAR estimation by the unprecedented volatility of the pandemic years, and contamination of the instrument itself by events that are pandemic-driven rather than genuine supply chain disruptions.

The first concern is already addressed in the baseline specification through the feasible-GLS procedure adapted from Lenza and Primiceri (2022) and Hamilton (Forthcoming), which downweights the pandemic observations by an estimated variance scaling factor. As an additional check, I compute the structural impulse responses after dropping the entire 2020 calendar year from the sample. The results are essentially indistinguishable from the baseline at every horizon (see Online Appendix B.1 Figure B.3). An alternative would be to restrict

the sample to the pre-2020 period, but only 17 of the 43 non-zero monthly observations of the instrument fall in that window. I therefore extend the GLS scaling procedure to cover the full 2020–2024 period, downweighting both the pandemic and the post-pandemic observations by the same factor in the VAR estimation as well as in the construction of the instrument. The impulse responses confirm the main findings: a global supply chain shock generates a persistent disruption to the global supply chain and stagflationary effects on consumer prices and industrial production, whose magnitudes and shapes are comparable with the baseline (see Online Appendix B.1 Figure B.4).

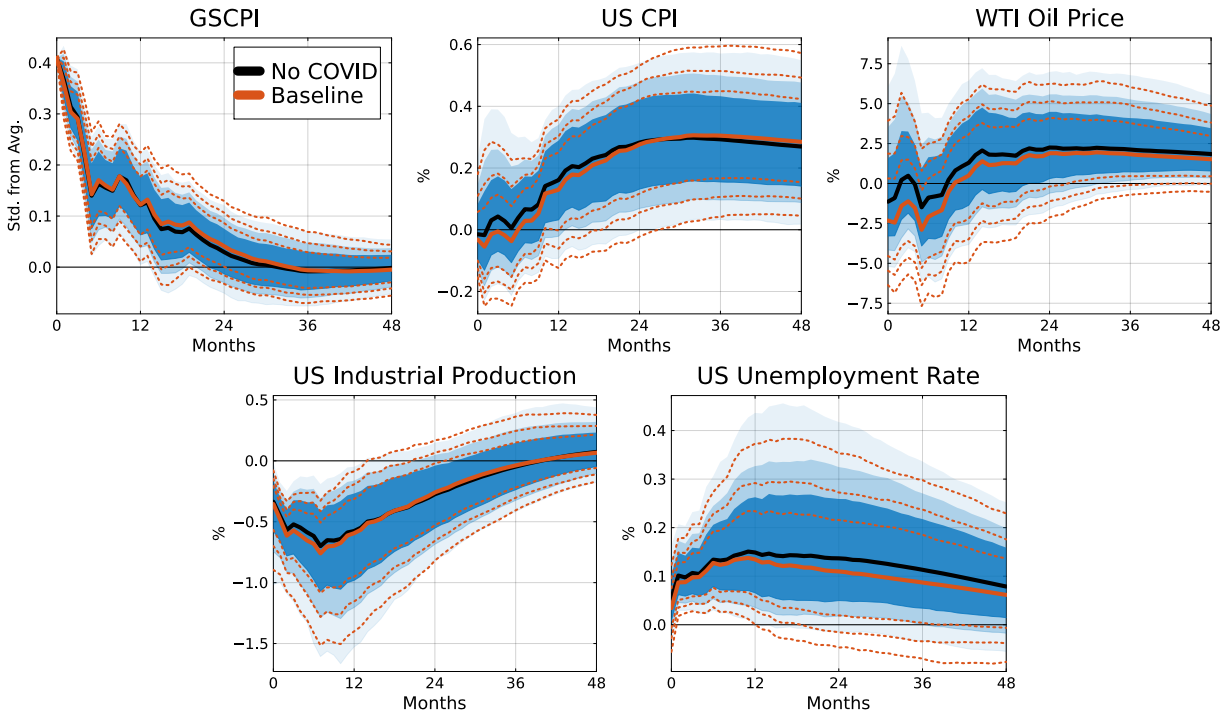


Figure 13: Robustness to Excluding COVID-Related Disruptions

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The orange lines represent the baseline specification, while the black lines represent the restricted estimates excluding the COVID-related disruptions. The solid lines represent the point estimates, and the shaded areas (and dotted lines) denote 68, 80, and 90 percent confidence bands.

I address the concern about the contamination of the instrument in two ways. Of the 54 events in the proxy, the narrative analysis confirms that only three are attributable to pandemic-related events.[6] The first event is the closure of the Meishan terminal at Ningbo decided by the Chinese authorities in August 2021 after a single dockworker tested positive:

[6] An earlier version of this paper, which did not include a systematic cross-checking of the price surcharge announcements, reported additional events related to COVID-19. Independent verification against external sources led to the removal or reclassification of most of those events, leaving only three for which the proximate cause is pandemic-related.

a shock that paralysed an entire facility rather than a true pandemic-driven shock. The other two are the introduction of secure-workplace protocols and other sanitary measures at Felixstowe in September 2020 and across all the United Kingdom container gateways in November 2020. Excluding these three events from the proxy yields the impulse responses reported in Figure 13, which are essentially unchanged relative to the baseline.

Taken together, dropping the 2020 observations from the estimation, downweighting the entire 2020–2024 window through the GLS scaling block, and removing the pandemic-related events from the instrument all leave the conclusions intact: the impulse responses are not driven by the pandemic period.

6.6 Testing the Invertibility Assumption

The identification strategy relies on the invertibility assumption, namely that the structural shocks at each horizon lie within the linear span of current and lagged values of the endogenous variables. I provide three pieces of evidence supporting the invertibility of the baseline VAR specification.

First, the nature of the identified shock makes invertibility ex ante plausible. Unlike news shocks, which Ramey (2016) and Nakamura and Steinsson (2018) argue are a common source of invertibility failures, the global supply chain shock does not involve a foresight problem; in addition, the GSCPI in the VAR is itself a global aggregator of disruption signals, which limits the scope for omitted information to drive a wedge between the structural shocks and the linear span of the data.

Second, I implement the invertibility test proposed by Plagborg-Møller and Wolf (2022) together with the joint variant of Hamilton (Forthcoming). Under the null of invertibility, lagged values of the instrument should add no predictive content to the VAR equations once the endogenous regressors are included. Equation-by-equation F-tests fail to reject the null in every VAR equation, and the joint test that pools the lags of the instrument across all five equations simultaneously fails to reject as well (see Online Appendix B.2 Table B.5).

Third, I relax the invertibility assumption explicitly using the proxy-SVAR methodology of Forni, Gambetti and Ricco (2022), which is invertibility-robust by construction. Their procedure first cleans the external instrument of any contemporaneous correlation with past values of itself and of other endogenous variables, and then identifies the structural impact effects from a regression of the cleaned proxy on the contemporaneous and lead reduced-form residuals obtained from the VAR. Figure 14 reports the impulse response functions for 216 different specifications of the invertibility-robust approach, obtained by varying the number of lags of the instrument and the endogenous variables used to clean the proxy from auto-

correlation, and the number of leads of the reduced-form residuals to ensure lead exogeneity. All the responses lie tightly on top of the baseline, confirming the robustness of the results with respect to the invertibility assumption. The authors also propose two diagnostic tests and both fail to reject: the first, the joint null that the lags of the instrument in the cleaning regression are zero, shows that the proxy is free of predictable past information; the second, the joint null that the leads of the reduced-form residuals in the lead regression are zero, shows that invertibility holds, since the proxy carries no information about future structural shocks. Neither null is rejected at conventional levels (see Online Appendix B.2 Table B.6). Invertibility is therefore consistent with the data, and even relaxing it using the approach of Forni, Gambetti and Ricco (2022) returns the same impulse responses.

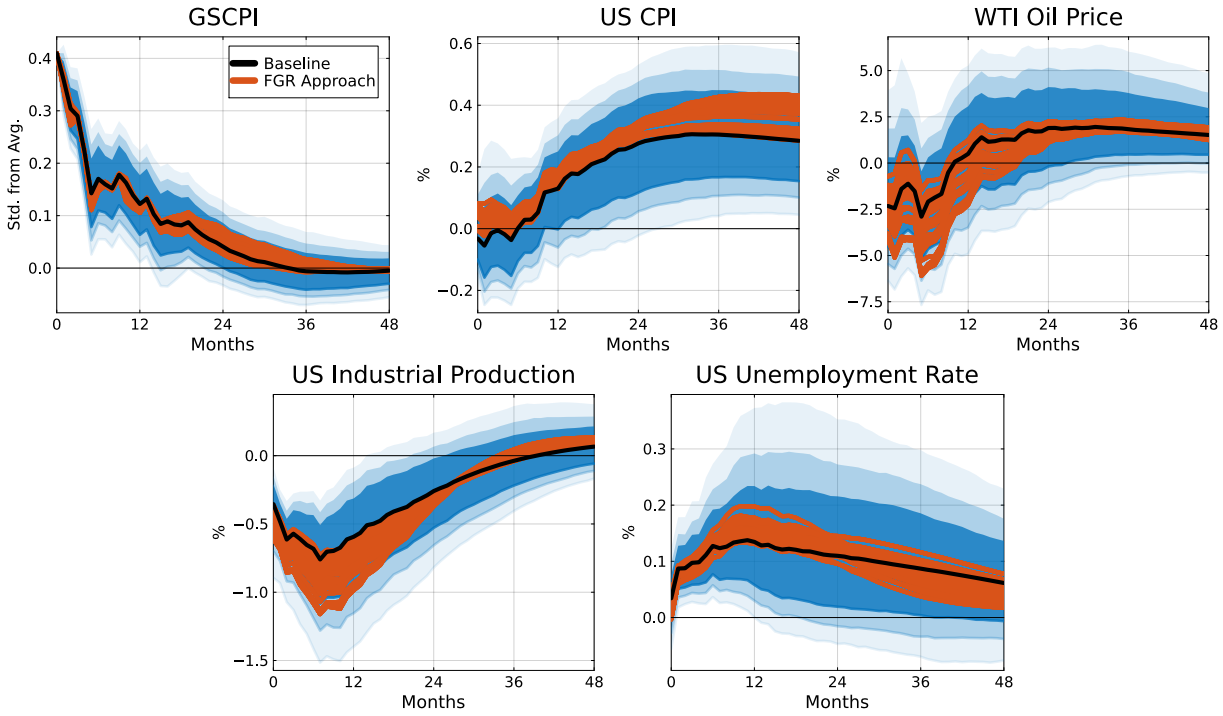


Figure 14: Invertibility-Robust Impulse Responses

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The black line is the baseline estimate, surrounded by 68, 80, and 90 percent confidence bands. The orange lines report the impulse responses from the Forni, Gambetti and Ricco (2022) procedure, computed over a $6 \times 6 \times 6$ grid for the number of lags of the instrument (m), the number of lags of the VAR variables (q), and the number of leads of the residuals (r).

6.7 Is the VAR Structure Misspecified?

A final concern is that the VAR may not adequately capture the dynamic propagation of the shock. As Montiel Olea et al. (2025) note, in a VAR(p) the long-run impulse responses

are extrapolated from the first p sample autocovariances, and a misspecified lag length can in principle distort the estimates at long horizons. However, estimating a separate Local Projection-IV regression at each horizon would be infeasible given the short instrument window and would severely curtail the effective sample size at long horizons. Instead, I follow the approach of Känzig (2021) and use the Proxy-SVAR-identified shock $\hat{u}_{1t}$ as the right-hand-side variable in a battery of local projections to compare the impulse responses obtained under and without the dynamic restrictions of the VAR:

$$\boldsymbol{y}_{t+h} \;=\; \boldsymbol{\Theta}_{h,1}\,\hat{u}_{1t} \;+\; \boldsymbol{\Pi}_h\,\boldsymbol{x}_{t-1} \;+\; \boldsymbol{\eta}_{t+h}\,, \tag{15}$$

where $\boldsymbol{y}_{t+h}$ denotes the vector of outcome variables at horizon h, and $\boldsymbol{x}_{t-1}$ is a vector of controls including twelve lags of the outcome variables.

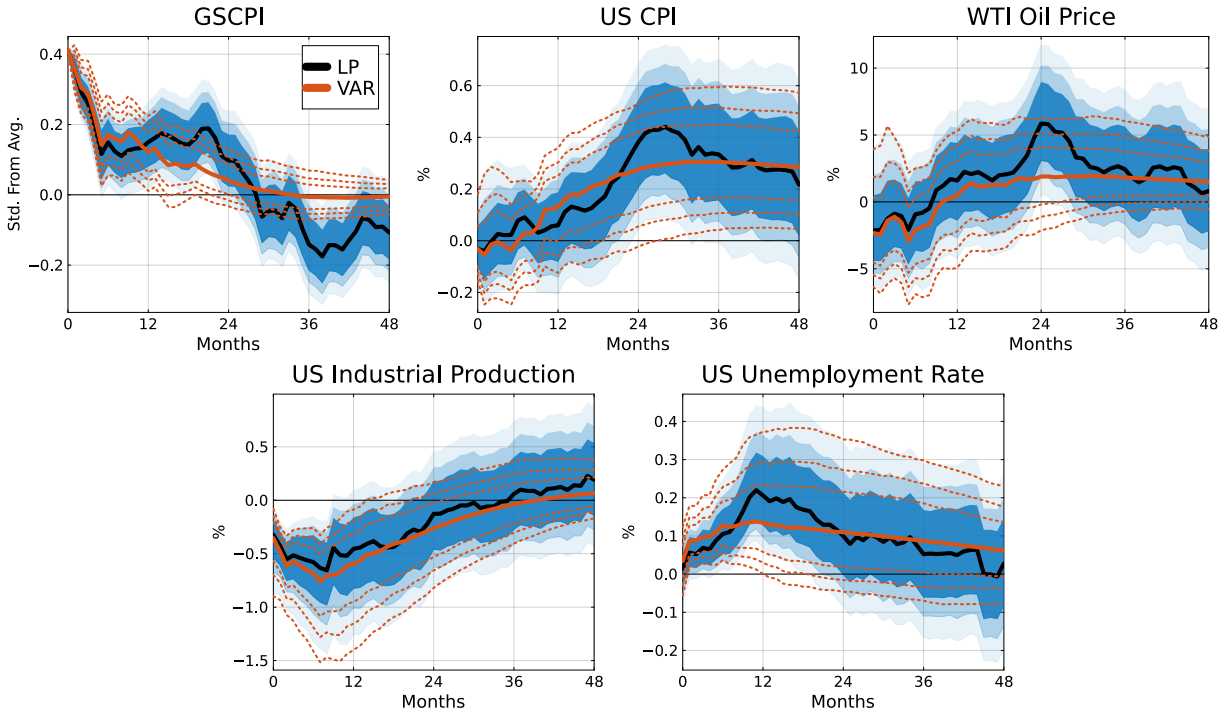


Figure 15: Proxy-SVAR vs. Local Projection Responses

Notes: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The orange lines represent the baseline Proxy-SVAR results, while the black lines represent the local-projection results obtained using the SVAR-identified shock. The solid lines represent the point estimates, and the shaded areas (and dotted lines) denote 68, 80, and 90 percent confidence bands.

Figure 15 shows that the responses are remarkably similar across both short and long horizons. As expected, the local projections are mechanically more erratic, but they trace out the same trajectories as the Proxy-SVAR.

I also conduct additional robustness exercises in Online Appendix B, where the baseline results are shown to be robust across alternative VAR lag orders, to the inclusion of 73 controls one at a time, and to estimating the VAR from 2013 onward, so that the estimation sample coincides with the coverage of the instrument.

7 Conclusion

This paper presents a novel identification strategy to assess the impact of shocks to the global supply chain on the macroeconomy. I identify 54 exogenous disruptions worldwide, derived from a narrative analysis of the 7,872 price and surcharge updates from the three largest container shipping companies. These shocks have significant aggregate effects on the macroeconomy and act as cost-push factors, raising prices persistently and exerting long-lasting downward pressure on economic activity.

I show that these shocks produce a broad-based contraction across all major production sectors and trigger persistent increases in consumer prices across nearly all major CPI subcategories. The magnitude of these effects varies systematically with each sector's exposure to the global supply chain, proxied by the percentage of production inputs outsourced abroad. I demonstrate that sectors with higher exposure experience sharper output contractions and stronger price pass-through. Beyond the tradable goods sector, I find that global supply chain shocks spill over into non-tradable services. Although the response of service activity and employment is more modest and transitory, it highlights the wider macroeconomic reach of the global supply chain shocks.

I show that negative shocks to the global supply chain were key drivers of inflation in the early stages of the post-pandemic period, accounting for an average of 45% of the increase in consumer prices in 2021. This contribution diminished over time but remained significant in 2022 (29%), 2023 (28%), and 2024 (21%). I also find that these shocks imposed a substantial drag on economic recovery. In a counterfactual scenario with only global supply chain shocks, industrial production would have returned to pre-pandemic levels 18 months later than it did.

I further demonstrate that global supply chain shocks are significant drivers of business cycle fluctuations, accounting for approximately 24% of the long-run variance in industrial production and 30% of the variance in consumer prices, thereby positioning them as key contributors to business cycle fluctuations.

I find that these results are robust across a wide range of empirical specifications, including alternative identification schemes, identification assumptions, and different specifications of the instrument. My identification strategy, centered on shipping company surcharges, offers a credible way to isolate exogenous supply-side disruptions.

References

Acemoglu, Daron, and Alireza Tahbaz-Salehi. 2020. "Firms, failures, and fluctuations: the macroeconomics of supply chain disruptions." National Bureau of Economic Research.

Acemoglu, Daron, and Alireza Tahbaz-Salehi. 2025. "The Macroeconomics of Supply Chain Disruptions." *Review of Economic Studies*, 92(2): 656–695.

Acemoglu, Daron, Vasco M Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. 2012. "The network origins of aggregate fluctuations." *Econometrica*, 80(5): 1977–2016.

Alessandria, George, Shafaat Yar Khan, Armen Khederlarian, Carter Mix, and Kim J Ruhl. 2023. "The aggregate effects of global and local supply chain disruptions: 2020–2022." *Journal of international Economics*, 146: 103788.

Alquist, Ron, Saroj Bhattarai, and Olivier Coibion. 2020. "Commodity-price comovement and global economic activity." *Journal of Monetary Economics*, 112: 41–56.

Aruoba, S Borağan, and Thomas Drechsel. Forthcoming. "Identifying monetary policy shocks: A natural language approach." *American Economic Journal: Macroeconomics*.

Ascari, Guido, Dennis Bonam, and Andra Smadu. 2024. "Global supply chain pressures, inflation, and implications for monetary policy." *Journal of International Money and Finance*, 142: 103029.

Bai, Xiwen, Jesús Fernández-Villaverde, Yiliang Li, and Francesco Zanetti. Forthcoming. "The causal effects of global supply chain disruptions on macroeconomic outcomes: evidence and theory." *American Economic Review*.

Baker, Scott R, Nicholas Bloom, and Steven J Davis. 2016. "Measuring economic policy uncertainty." *The Quarterly Journal of Economics*, 131(4): 1593–1636.

Baldwin, Richard, Rebecca Freeman, and Angelos Theodorakopoulos. 2023. "Hidden exposure: Measuring US supply chain reliance." *Brookings Papers on Economic Activity*, 2023(2): 79–167.

Baqaee, David, and Emmanuel Farhi. 2022. "Supply and demand in disaggregated Keynesian economies with an application to the Covid-19 crisis." *American Economic Review*, 112(5): 1397–1436.

Barrot, Jean-Noël, and Julien Sauvagnat. 2016. "Input specificity and the propagation of idiosyncratic shocks in production networks." *The Quarterly Journal of Economics*, 131(3): 1543–1592.

Baumeister, Christiane, and James D Hamilton. 2019. "Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks." *American Economic Review*, 109(5): 1873–1910.

Baumeister, Christiane, and James D Hamilton. 2024. "Advances in using vector autoregressions to estimate structural magnitudes." *Econometric Theory*, 40(3): 472–510.

Baumeister, Christiane, Dimitris Korobilis, and Thomas K Lee. 2022. "Energy markets and global economic conditions." *Review of Economics and Statistics*, 104(4): 828–844.

Benigno, Gianluca, Julian Di Giovanni, Jan J Groen, and Adam I Noble. 2022. "The GSCPI: A new barometer of global supply chain pressures." FRB of New York Staff Report 1017.

Bergholt, Drago, Fabio Canova, Francesco Furlanetto, Nicolò Maffei-Faccioli, and Pål Ulvedal. Forthcoming. "What drives the recent surge in inflation? The historical decomposition roller coaster." *American Economic Journal: Macroeconomics*.

Bernanke, Ben, and Olivier Blanchard. 2023. "What caused the US pandemic-era inflation?" *PIIE, Peterson Institute for International Economics*, 86.

Bernanke, Ben, and Olivier Blanchard. 2024. "An Analysis of Pandemic-Era Inflation in 11 Economies." National Bureau of Economic Research.

Boehm, Christoph E, Aaron Flaaen, and Nitya Pandalai-Nayar. 2019. "Input linkages and the transmission of shocks: Firm-level evidence from the 2011 Tōhoku earthquake." *Review of Economics and Statistics*, 101(1): 60–75.

Brinca, Pedro, Joao B Duarte, and Miguel Faria-e Castro. 2021. "Measuring labor supply and demand shocks during COVID-19." *European Economic Review*, 139: 103901.

Caldara, Dario, Matteo Iacoviello, and David Yu. 2025. "Measuring shortages since 1900." International Finance Discussion Paper 1407.

Caldara, Dario, Michele Cavallo, and Matteo Iacoviello. 2019. "Oil price elasticities and oil price fluctuations." *Journal of Monetary Economics*, 103: 1–20.

Carrière-Swallow, Yan, Pragyan Deb, Davide Furceri, Daniel Jiménez, and Jonathan D Ostry. 2023. "Shipping costs and inflation." *Journal of International Money and Finance*, 130: 102771.

Carvalho, Vasco M, Makoto Nirei, Yukiko U Saito, and Alireza Tahbaz-Salehi. 2021. "Supply chain disruptions: Evidence from the great east japan earthquake." *The Quarterly Journal of Economics*, 136(2): 1255–1321.

Cochrane, John H. 2022. "Inflation past, present and future: Fiscal shocks, Fed response, and fiscal limits." National Bureau of Economic Research.

Comin, Diego A, Robert C Johnson, and Callum J Jones. 2023. "Supply chain constraints and inflation." National Bureau of Economic Research.

Davis, Steven J, and John Haltiwanger. 2001. "Sectoral job creation and destruction responses to oil price changes." *Journal of monetary economics*, 48(3): 465–512.

Delle Chiaie, Simona, Laurent Ferrara, and Domenico Giannone. 2022. "Common factors of commodity prices." *Journal of Applied Econometrics*, 37(3): 461–476.

den Besten, Tamar, Regis Barnichon, Diego R Känzig, and Aayush Singh. 2026. "The Macroeconomic Effects of Tariffs: Insights from 180 Years of US Trade Policy." National Bureau of Economic Research.

De Santis, Roberto A. 2024. "Supply chain disruption and energy supply shocks: impact on euro area output and prices."

Di Giovanni, Julian, Şebnem Kalemli-Özcan, Alvaro Silva, and Muhammed A Yildirim. 2022. "Global supply chain pressures, international trade, and inflation." National Bureau of Economic Research.

Di Giovanni, Julian, Şebnem Kalemli-Özcan, Alvaro Silva, and Muhammed A Yildirim. 2023*a*. "Pandemic-era inflation drivers and global spillovers." National Bureau of Economic Research.

Di Giovanni, Julian, Şebnem Kalemli-Özcan, Alvaro Silva, and Muhammed A Yıldırım. 2023*b*. "Quantifying the inflationary impact of fiscal stimulus under supply constraints." *AEA Papers and Proceedings*, 113: 76–80.

Elliott, Matthew, and Benjamin Golub. 2022. "Networks and economic fragility." *Annual Review of Economics*, 14(1): 665–696.

Elliott, Matthew, Benjamin Golub, and Matthew V Leduc. 2022. "Supply network formation and fragility." *American Economic Review*, 112(8): 2701–2747.

Fan, Lei, William W Wilson, and Bruce Dahl. 2012. "Congestion, port expansion and spatial competition for US container imports." *Transportation Research Part E: Logistics and Transportation Review*, 48(6): 1121–1136.

Faria-e Castro, Miguel. 2024. "The St. Louis Fed DSGE Model." FRB of St. Louis Working Paper.

Finck, David, and Peter Tillmann. 2022. "The macroeconomic effects of global supply chain disruptions." *BOFIT Discussion Paper*.

Forni, Mario, Luca Gambetti, and Giovanni Ricco. 2022. "External instrument SVAR analysis for noninvertible shocks."

Gabaix, Xavier. 2011. "The granular origins of aggregate fluctuations." *Econometrica*, 79(3): 733–772.

Garcia Revelo, Jose David, Gregory Levieuge, and Jean-Guillaume Sahuc. Forthcoming. "Revisiting 15 Years of Unusual Transatlantic Monetary Policies." *Journal of International Money and Finance*.

Giannone, Domenico, and Giorgio Primiceri. 2024. "The drivers of post-pandemic inflation." National Bureau of Economic Research.

Goda, Gopi Shah, and Evan J Soltas. 2022. "The impacts of Covid-19 illnesses on workers." *Journal of Public Economics*, 222.

Gordon, Matthew V, and Todd E Clark. 2023. "The impacts of supply chain disruptions on inflation." Economic Commentary, Federal Reserve Bank of Cleveland 2023-08.

Guerrieri, Veronica, Michala Marcussen, Silvana Tenreyro, and Lucrezia Reichlin. 2023. *The art and science of patience: relative prices and inflation.* Centre for Economic Policy Research (CEPR).

Hamilton, James D. 1985. "Historical causes of postwar oil shocks and recessions." *The Energy Journal*, 6(1): 97–116.

Hamilton, James D. 2003. "What is an oil shock?" *Journal of econometrics*, 113(2): 363–398.

Hamilton, James D. 2009. "Causes and Consequences of the Oil Shock of 2007-08." *Brookings Papers on Economic Activity*.

Hamilton, James D. Forthcoming. *Vector Autoregressions*. Cambridge University Press.

Hazell, Jonathon, and Stephan Hobler. 2024. *Do Deficits Cause Inflation?: A High Frequency Narrative Approach.* London School of Economics and Political Science.

Herrera, Ana María, Latika Gupta Lagalo, and Tatsuma Wada. 2011. "Oil price shocks and industrial production: Is the relationship linear?" *Macroeconomic dynamics*, 15(S3): 472–497.

Hobijn, Bart, and Aysegul Sahin. 2022. ""Missing" Workers and "Missing" Jobs Since the Pandemic." FRB of Chicago Working Paper 2022-54.

Jentsch, Carsten, and Kurt G Lunsford. 2019. "The dynamic effects of personal and corporate income tax changes in the United States: Comment." *American Economic Review*, 109(7): 2655–2678.

Jordà, Òscar, and Fernanda Nechio. 2023. "Inflation and wage growth since the pandemic." *European Economic Review*, 156: 104474.

Känzig, Diego R. 2021. "The macroeconomic effects of oil supply news: Evidence from OPEC announcements." *American Economic Review*, 111(4): 1092–1125.

Känzig, Diego R, and Ramya Raghavan. 2026. "Supply Chain Shocks and the Macroeconomy: Evidence from Global Shipping Disruptions." Working paper, available at https://dkaenzig.github.io/diegokaenzig.com/Papers/kr_supplychainshocks.pdf.

Lane, Philip. 2022. "Bottlenecks and monetary policy." *ECB blog*.

Lee, Dain, Jinhyeok Park, and Yongseok Shin. 2023. "Where are the workers? From great resignation to quiet quitting." National Bureau of Economic Research.

Lenza, Michele, and Giorgio E Primiceri. 2022. "How to estimate a vector autoregression after March 2020." *Journal of Applied Econometrics*, 37(4): 688–699.

Mertens, Karel, and Morten O Ravn. 2013. "The dynamic effects of personal and corporate income tax changes in the United States." *American economic review*, 103(4): 1212–1247.

Montiel Olea, José Luis, and Carolin Pflueger. 2013. "A robust test for weak instruments." *Journal of Business & Economic Statistics*, 31(3): 358–369.

Montiel Olea, José Luis, James H Stock, and Mark W Watson. 2021. "Inference in structural vector autoregressions identified with an external instrument." *Journal of Econometrics*, 225(1): 74–87.

Montiel Olea, José Luis, Mikkel Plagborg-Moller, Eric Qian, and Christian Wolf. 2025. "Local Projections or VARs? A Primer for Macroeconomists." NBER Working Paper w33871.

Nakamura, Emi, and Jón Steinsson. 2018. "Identification in macroeconomics." *Journal of Economic Perspectives*, 32(3): 59–86.

Notteboom, Theo, Athanasios Pallis, and Jean-Paul Rodrigue. 2022. *Port economics, management and policy.* Routledge.

Plagborg-Møller, Mikkel, and Christian K Wolf. 2022. "Instrumental variable identification of dynamic variance decompositions." *Journal of Political Economy*, 130(8): 2164–2202.

Qureshi, Irfan A, and Ghufran Ahmad. 2025. "Oil price shocks and US business cycles." *Journal of Economic Dynamics and Control*, 105132.

Ramey, Valerie A. 2011. "Identifying government spending shocks: It's all in the timing." *The Quarterly Journal of Economics*, 126(1): 1–50.

Ramey, Valerie A. 2016. "Macroeconomic shocks and their propagation." *Handbook of macroeconomics*, 2: 71–162.

Ramey, Valerie A, and Matthew D Shapiro. 1998. "Costly capital reallocation and the effects of government spending." *Carnegie-Rochester conference series on public policy*, 48: 145–194.

Romer, Christina D, and David H Romer. 2004. "A new measure of monetary shocks: Derivation and implications." *American economic review*, 94(4): 1055–1084.

Romer, Christina D, and David H Romer. 2010. "The macroeconomic effects of tax changes: estimates based on a new measure of fiscal shocks." *American economic review*, 100(3): 763–801.

Rubbo, Elisa. 2024. "What drives inflation? Lessons from disaggregated price data." National Bureau of Economic Research.

Shapiro, Adam Hale. 2024. "Decomposing Supply-and Demand-Driven Inflation." *Journal of Money, Credit and Banking*.

Stock, James H. 2008. "What's new in Econometrics: Time Series, lecture 7." *Short course lectures, NBER Summer Institute*.

Stock, James H, and Mark W Watson. 2016. "Dynamic factor models, factor-augmented vector autoregressions, and structural vector autoregressions in macroeconomics." In *Handbook of macroeconomics*. Vol. 2, 415–525. Elsevier.

Stock, James H, and Mark W Watson. 2018. "Identification and estimation of dynamic causal effects in macroeconomics using external instruments." *The Economic Journal*, 128(610): 917–948.

Tenreyro, Silvana. 2021. "International trade, global supply chains and monetary policy." *Speech given at the Centre for Economic Policy Research webinar*.

Tenreyro, Silvana. 2022. "The path to 2 percent." *Speech at the Society of Professional Economists Annual Conference, London*.